

School of Electronic Engineering
and Computer Science

Final Report

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BEng Electrical and Electronic
Engineering

Project Title:

**High Altitude Balloon
to take Near Space
Measurements**

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ABSTRACT

One of the challenges of high altitude exploration is that there are limited ways of doing so without spending vast sums of money. However, high altitude ballooning was developed as a means to explore the first two layers of the atmosphere effectively. This method of exploration is subject to strict requirements, and throughout this project measures are taken to ensure that these are adhered to. These regulations form the basis of a transmission system which is capable of transmitting data from the maximum altitude of the balloon to receivers on the ground below.

A range of experiments were designed, including temperature sensing, humidity sensing and solar power measurement. Care was also taken when designing the payload to ensure that it was structurally sound and robust. A recovery mechanism in the form of a parachute was also designed and tested.

To design the transmission system, two different types of antennas were researched, simulated and tested to ensure that correct operation. A modulation and data encoding system was researched and implemented to provide the live data transmission. A receiver was also tested and provided reasonable signal to noise ratio results. Finally, the two antennas and the payload were tested. The payload showed a pre-flight temperature change of only 5 degrees for a 40 degree temperature variation. The patch antenna provided -22dB return loss and a bandwidth of 3%. The Yagi antenna provided 9dBi of gain at 434MHz. With these results, the payload and transmission system was ready to be used.

The launch of the balloon took place on the 20th April 2013 in Ellsworth, Cambridge, and the balloon was tracked for two hours, before landing in a field 30 miles from the launch site. The data recorded throughout the flight is presented in this report. The temperature variations recorded are consistent with previous experimental data. The transmission system was so successful that the balloon was able to be tracked from Yorkshire. Adding to this, the GPS location system pinpointed an accurate landing position for the balloon, allowing it to be recovered.

Further improvements to the system are suggested throughout and emphasized in the concluding chapter. The objectives set out in the initial chapters have all been met through the careful design, construction and testing of the payload.

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CHAPTER 1: INTRODUCTION

1.1 CONTEXT

It was in September 1892 that a French aeronaut named Gustave Hermite launched the first sounding balloon made of waxed paper and consisting of a barometer. (Lynch, 2006) .Three years later, meteorologist, Leon Phlippe Teisserenc de Bort, frustrated at the lack of funding in his current employment, resigned, to pursue his own research of the atmosphere. The method he used was the same method that Hermite pioneered in 1892. De Bort was so dedicated that by the point his work was presented to the French Academy of Sciences, he had launched 236 balloons to further his studies of the atmosphere. It was he who proposed that the atmosphere was composed of several separate zones, giving each a name, putting forward new radical theories, and proving them through the use of his balloon flights. (Allen, 1983)

Throughout the remainder of the 20th century, the study of the atmosphere continued, leading to development of communication systems based around the electromagnetic properties of the atmosphere, weather predictions and pollution measurement. However, the methods of research have not changed over the past 100 years, and high altitude balloons are still a primary method for meteorological research.

Now, despite almost every aspect of the atmosphere being measured, analysed and observed, there are still many unanswered questions, more experiments to be carried out, and new ways to understand and use what is known about the atmosphere to make steps forward in research.

1.2 MOTIVATION

Little is known about a vast areas of the atmosphere, so little that researchers often refer to the region above the stratosphere at above 50km as the “*ignorosphere*” (Siskind, et al., 2000) This is due to inaccessibility by balloons sent from the ground, and by satellites in low earth orbit. A great deal of research is being carried out both by climatologists and meteorologists to understand the processes of the earth’s atmosphere. This is both for daily forecasting purposes and for research into the long term effects of the changes in the atmosphere.

As the two lower regions of the atmosphere are most accessible, there are less gaps in research and less to be improved than in other areas of the atmosphere. However new technology such as Heliosats which propose power transmission from the stratosphere, represent new areas of research and improvement. The primary issue to address here however, is what is the current state of the atmosphere and how can information gained about it be used to further understand it.

1.3 AIMS AND OBJECTIVES

The overall aim of this project is to launch and recover a research balloon that will ascend to at least 60,000 feet with a payload of various different sensors and recording equipment. The payload should consist of an experiment to determine the optimum solar panel tilt at different altitudes to determine if this is needed for future projects.

This project has a number of objectives to achieve:

- Research and develop a live data transmission system using a radio transceiver system
- Design a solar angle optimum tilt experiment
- Design an insulated payload with possible back up heating system and suitable parachute to enable recovery.
- Calibrate and test the sensors and data equipment prior to launch.
- Record imagery of the whole balloon flight.
- Take measurements of the earth's atmosphere at heights above 60,000 feet using calibrated sensors.
- Determine if it is necessary to include solar tilting mechanisms for future projects or large scale, high altitude photovoltaic systems.

1.4 REPORT APPROACH.

The final report will have a structure as follows:

Chapter1: Introduction – In this section a very brief historical context is given with the aims and objectives outlined.

Chapter2: Background research – this will contain a literature review which will focus on the experiments that a research balloon can carry and the alternative methods for exploring the upper atmosphere.

Chapter 3: Requirements – this will be a brief chapter stating the requirements of the balloon based upon the research conducted in Chapter 2 and the objectives from Chapter 1, combined with the regulatory requirements for which the system must meet.

Chapter 4: Instrumentation & Payload: Research, Design and Implementation – this section includes the development of the instruments which will record data throughout the balloon flight. This data must be transmitted from the payload, and the methodology and development of the system for this task is included in the next chapter. The final circuit design is also included at the end of this chapter with the exception of any hardware for the transmitter circuit.

Chapter 5: Transceiver System: Research, Design and Implementation – This section firstly includes the research pertaining to the FM modulation system, followed by the practical implementation of it in terms of the hardware and software required. Following this, a research based comparison between the various methodologies of implementing the payload and ground antennas is carried out. The choices made here are then implemented and developed using CAD software. The CAD design is then prepared for fabrication, and a number of tuning methods developed.

Chapter 6: Fabrication development, Testing and analysis – This chapter includes the fabricated payload antenna development, the final results for it and a comparison between these results and the CAD results. There will also be the fabricated ground antenna results, analysed in a similar fashion. In this chapter the payload is fabricated, developed and tested.

Chapter 7: In-flight Results Analysis – this will show the results of the flight and will also contain a comparison between other experimental results and the results obtained in this flight.

Chapter 8: Conclusion – in this section the various aspects of the project will be critically analysed to find areas for improvements and for possible future study.

CHAPTER 2: BACKGROUND RESEARCH

As time and technology progresses, the ability to measure and experiment at higher altitudes becomes more and more possible, and widely available. As a result of this there have been many studies on the atmosphere to measure the current climate of the Earth in relation to pollution, the properties of the sun, and the general cycles that the planet goes through. It is important to look into these experiments and find where there may be shortcomings in the data or areas to improve. By first looking at the structure of the atmosphere and methods of transport for recording equipment, a viable option can be chosen to gather data. Once this is chosen, the specific systems and experiments that will be carried out can be defined by critical analysis of previous work and data.

2.1 THE EARTH'S ATMOSPHERE

The structure of the atmosphere was first defined by De Bort, however, his initial findings had some inaccuracies in the ranges of each zone. Nowadays, the atmosphere is defined by 7 distinct zones as outlined in (Allen, 1983). The atmosphere is an extremely vast and relatively unexplored area. The troposphere is the area closest to the surface of the earth, and the area in which all living things are located. Even then, only the lowest parts are habitable – the rest of the atmosphere is exposed to radiation and extreme temperatures. Above the 10 mile reach of the troposphere, the stratosphere is exposed to ultraviolet radiation, which in fact causes a temperature rise. Above this, in the mesosphere, the temperatures are lower, and at its highest reaches drop to hundreds of degrees below zero. Another temperature inversion is seen in the thermosphere where solar energy heats up the air to a few thousand degrees above zero. (Allen, 1983). The temperature inversions present are often recorded on balloon flights and this can be seen in Figure 1. (Larson, et al., 2009)

From sea-level to the upper limit of the stratosphere at 40km high, information can be gathered using high altitude balloons. To measure above this point, any equipment must be sent with some form of propulsion; such experiments require funding, significant research and time. The region above the 150km mark can often be analysed by orbiting

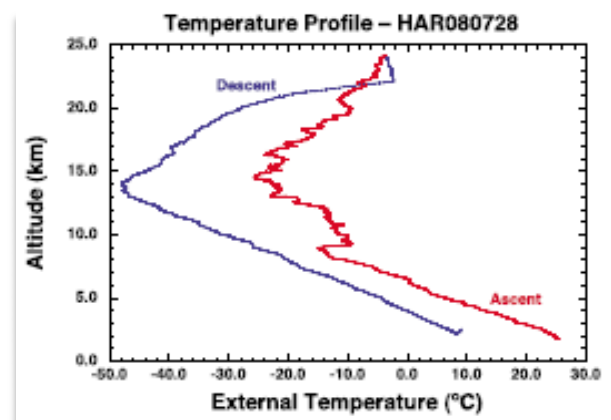


FIGURE 1: TEMPERATURE WITH RESPECT TO ALTITUDE OVER A HIGH ALTITUDE BALLOON FLIGHT (IMAGE FROM LARSON ET.AL. 2009)

satellites, however between the upper edge of the stratosphere and the lower reaches of these satellites, experimentation is difficult. According to (Siskind, et al., 2000) this is now less of a problem than it was previously as techniques such as remote sensing are available to study the regions from the mesosphere upwards.

Despite all the technology available, with a small budget of time and money it is not possible to expand knowledge of the relatively unknown mesosphere. In a tradeoff between budget and altitude, the upper reaches of the stratosphere at 40km high can be studied by the use of high altitude balloons. Here, the changes in the atmosphere cause the changes in the weather on the ground below. With the changes in the climate on the ground such as higher temperatures in the Polar Regions, rising sea levels and pollution, exploring the lower levels of the atmosphere is becoming more and more important and necessary.

2.2 EXPLORING THE TROPOSPHERE AND STRATOSPHERE.

The two lower parts of the earth's atmosphere are perhaps the most important. It is in the troposphere where clouds are formed, and thus a lot of our weather changes initiate here. Above the troposphere, the upper reaches of the stratosphere contain the ozone layer. This layer plays a crucial part in understanding temperature changes throughout the globe. This is particularly prominent in The Antarctic, where data collected from previous high altitude balloon flights over the past 30 years showed an increasing temperature change. This in turn correlated with a decrease in sea ice concentration over the same period. (Turner, et al., 2005)

With new technology available, there are other methods to profile and record data in the atmosphere. (Anthes, 2011) uses a method known as radio occultation. The basic method requires at least two satellites: a GPS satellite and Low Earth Orbiter (LEO) satellite. As the LEO satellite sets behind the earth, the 'bending' of the radio waves between the two satellites can be determined, thus obtaining a profile of the atmosphere based upon the angles and phase changes in the signals. This method provides a much more comprehensive view of the whole atmosphere compared with high altitude balloons. But it is clear that the results obtained show similarities between the two methods. What is also evident is the cost of equipment involved in this technique, showing once again that high altitude balloons provide an excellent altitude-for-money resource.

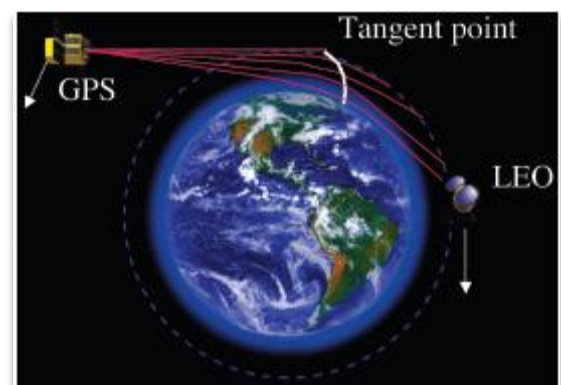


FIGURE 2: RADIO OCCULTATION OF THE ATMOSPHERE
(IMAGE SOURCED FROM ANTHERS, 2011)

2.3 HIGH ALTITUDE BALLOON SYSTEMS AND EXPERIMENTS

Having established that the best way to take data about the atmosphere without spending large sums of money is to use a high altitude balloon, it is important to look at what can be measured, how it can be measured, and what others have previously carried out.

2.3.1 EXPERIMENTS

(Dunlop, 2008) describes how weather balloons are used extensively in forecasting, with the most simple being just a balloon, with no payload to determine wind speed and direction. More

complex balloons are used, each carrying a radiosonde which is launched four times everyday across the world, taking accurate measurements for forecasting purposes. These carry a basic array of instruments including Global Positioning Satellite tracking, temperature, pressure and humidity. This is useful for forecasting; however, there are more experiments that can be carried to high altitudes to study the atmosphere.

Particle pollution is an important factor in health studies and in pollution measurement. In one year there are approximately 1300 million tons of particle pollutants discharged into the atmosphere. (Allen, 1983) Balloon borne equipment can collect stratospheric particulates by means of a dustsonde or by allowing particulates to impact on copper or carbon films. (Cawse & Tait, 1982) The main limitation in this instance is that this would only provide a snapshot of that area of the atmosphere, and due to the nature of pollution, this would not be conclusive as a study of general pollution in the atmosphere. Adding to this, injected particle pollutants which enter the stratosphere can remain there for years. This is due to updrafts in the atmosphere forcing the particles upwards meaning that they stay aloft longer than those that enter the troposphere (Allen, 1983)

The simplest experiment that can be done is to attach a camera to the balloon. This can then be advanced into multispectral imaging, providing imagery for a wide variety of purposes. High altitude balloons with cameras can be used by geologists to study rock formations, y those interested in studying pollution and by those studying heat using thermal imagery.

2.3.2 SYSTEMS

In order for the balloon to be recovered and tracked, most launches transmit GPS location strings through an antenna. (Beukers & Poppe, 1973) describes and designs a monopole style antenna specifically for a weather balloon. This is the most commonly used style of design in high altitude balloons, however, it has quite a large profile. A microstrip patch antenna could provide similar results if it was designed to be circularly polarized (to compensate for rotation in the payload). In this way, the antenna has a very low profile, whilst providing good transmission to the ground. A second type of antenna which could be an alternative to a monopole is the Helical antenna. This antenna serves as a circular polarized broadband antenna. According to (Rudge, et al., 1982) detailed design specifications for this type of antenna are limited. This means that the design process revolves mostly around computer simulation. Both the patch and helical antennae would be viable alternatives to the commonly used monopole antenna.

Combined with the antenna is the GPS system. Most GPS devices output a standard set of string formats known as National Maritime Electronics Association 0183 (NMEA). This is further expanded upon in the design section, however a useful guide is found in (El-Rabbany, 2002). By using this, balloons can be tracked very simply using a system which receives GPS data from the satellites, transmits through an FM transmitter and is received on the ground. The downfall of this method is that the transmitting antenna which must be robust otherwise recovery of a balloon would be difficult.

(Aglietti, et al., 2009) describes and tests the process of collection of solar energy using a high altitude platform. The experimental results show that the system could collect up to six times the

energy that a system on the ground could collect. The paper derives and calculates detailed equations for the Solar Zenith angle, which is dependent upon the air mass crossed by a light beam, the atmospheric thickness and path lengths in the earth's atmosphere. The angle in relation to the altitude and position is critical to obtain the maximum amount of solar energy collected. The equations derived in Aglietti et.al. would help calculate the angle at one particular altitude, and would require recalculation as the altitude increases. A system can be designed which, rather than calculating the angle, would analyse the output power from a scan across a range of angles and readjust the position to the optimum tilt based upon the output power. This would provide a simple optimisation of the tilt angle, without constantly implementing formulae at various altitudes.

2.4 RESEARCH SUMMARY

Despite the large amount of balloons launched in the last century, there is still room for improvement on many designs. Firstly, the communications, more specifically the antenna, is the primary area that can be modified. Two designs have been proposed in the research: the circular polarized patch antenna and the helical antenna. Both of these are viable alternatives and can be developed and simulated. These results will then be summarised and a comparison made to determine which is the most suitable for a high altitude balloon. In terms of experimentation, the main area for development is the solar tilt mechanism. This is an experiment which would have an application in the future, so the results may be quite important. Adding to this, there will be temperature, humidity and location measurements which will be compared to existing measurements.

CHAPTER 3: REQUIREMENTS

3.1 TRANSMISSION REQUIREMENTS

The Electromagnetic Spectrum is a heavily congested resource, and is widely used for Avionics, Business, Medical, Military and other public service uses. As such, it is in need of stringent control mechanisms to ensure that portions of the spectrum are allocated to certain services and uses, in turn providing a protection against interference from other frequency bands. In the United Kingdom the EM spectrum is allocated by the Office of Communications (Ofcom) and the Ministry of Defense (MOD). In this case, it is necessary to look into the regulations surrounding the use of transmitters in the public domain, particularly in terms of the licensing requirements.

Ofcom provides documentation showing that there are certain frequencies, which combined with transmission power restrictions, provide a license free option for transmission. This is beneficial since licensing can be expensive and time consuming. The regulations define a certain number of bands of frequencies ranging from 6.78MHz to 245GHz for use in "Non Specific Short Range Devices". UK Interface Requirements 2030 defines the precise regulations of the license exempt frequencies (Ofcom, 2011). The most commonly used of the frequency bands available are the 433.05-434.79 band, the 868-870MHz band and the 2.4GHz band. The final of these three is commonly used for microwave ovens and wireless LANs, meaning there may be some interference in this band. Any of these bands would be suitable for radio transmission from the payload to

ground, however the 434MHz band is commonly used for High Altitude Balloons, particularly when only data needs to be transmitted.

The regulations not only limit the frequency, but also limit the Equivalent Isotropically Radiated Power (EIRP). In the 434MHz band, the EIRP is limited to 10mW, whilst in the 869.40 – 869.65 MHz band, the Effective Radiated Power (ERP) permitted is up to 500mW. The transmitter modules for this frequency are slightly more expensive and the regulations also require a duty cycle of less than 10%. As such, the transmitter that will be used will be one which can operate in the 433.05 to 434.79 band.

3.2 PAYLOAD STRUCTURAL REQUIREMENTS

In order to launch a balloon, a number of structural requirements must be adhered to in addition to the transmission requirements. These have been summarised by the UK High Altitude Society and the relevant requirements have been outlined in this section

Firstly, to keep in line with the International Civil Aviation Organization (ICAO) regulations, the payload weight must be below 4kg in total. Those that are above 2kg, must have an area density of less than 13g per square centimeter.

In addition to weight, other requirements such as air traffic notifications are of vital importance. A Notice to Airmen (NOTAM) must be given for the launch site, and permission from the Civil Aviation Authority (CAA) must be obtained for the launch site and the balloon. It is possible to make use of a CAA exempt site, which makes the process easier.

The aim is to design the payload to be below 1kg (including parachute and balloon) which not only satisfies the regulations but ensures that the target altitude can be reached. Adding to this, a CAA exempt site is to be used to allow the focus to be maintained on the research and design of the various aspects of the payload.

CHAPTER 4: INSTRUMENTATION & PAYLOAD: RESEARCH, DESIGN AND IMPLEMENTATION

4.1 OVERVIEW AND DESIGN PROCEDURE

The project design has been split into two separate stages. This chapter includes the first stage: the instrumentation and the payload design. In this chapter, the various sensors and actuators will be outlined and developed. For most cases, the circuits will be designed using CAD software, then being implemented on breadboard, and finally on veroboard. Modifications may need to be made to the circuits during the implementation section, and these are documented in this section. The software which will allow the sensors and actuators to operate will be outlined in flow charts or in a textual explanation. The final code is available in the supporting material. Also in this section, the structure of the payload will be considered and developed.

The second section is included in chapter 5, and will document the process of designing the transceiver system. This section has been treated separately from the hardware and software of the payload as it requires its own research and has complex design considerations. It will include the

software and hardware for the transmitter and the antennas for both transmitting and receiving the data.

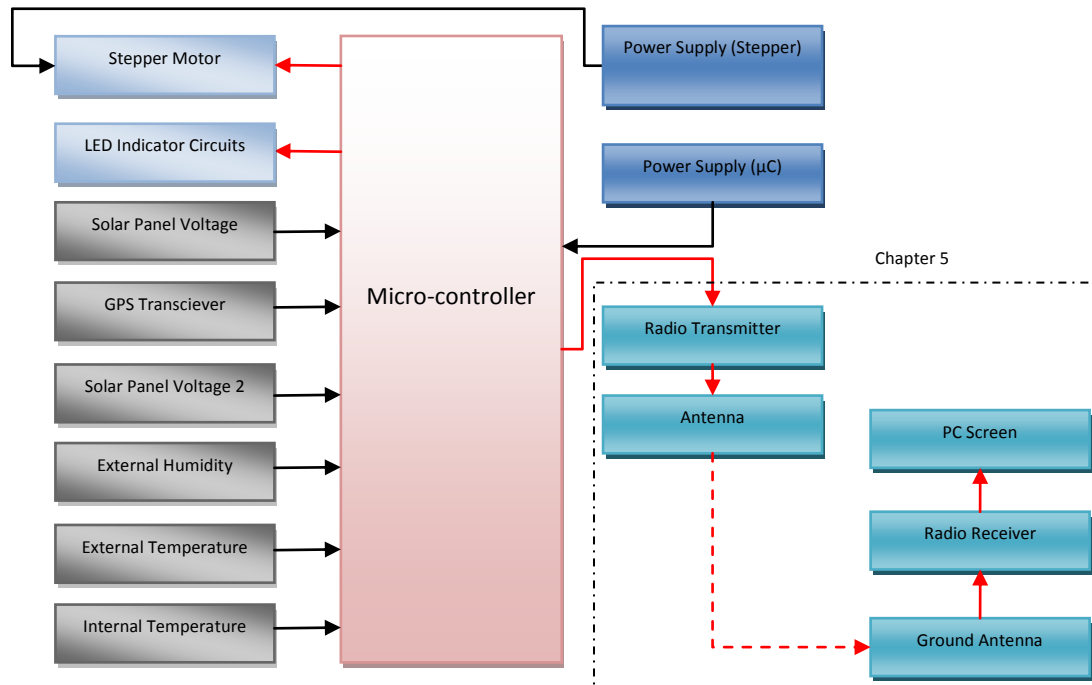


FIGURE 3: HIGH LEVEL BLOCK DIAGRAM OF OVERALL SYSTEM

Figure 3 shows how the microcontroller is the central element of the whole system. It takes inputs from the sensors and provides the signals to drive the actuators. It also provides the first stage of the modulation for the transmission of the data from the payload to the ground.

4.2 MICROCONTROLLER SELECTION

The central element to any automated system is the microcontroller. Microcontrollers provide great flexibility and reduce the cost and time of automating a system. By making use of a microcontroller, the design of the hardware can be focused away from ADCs and DACs and the complex issues which come with these systems, and placed on the software, external sensors and the actuators which make up the automated system itself (Deshmukh, 2005)

In order to best select the correct microcontroller for this task, the predicted system requirements based on the sensors and actuators should be detailed. The number of actuators will be limited, and will only consist of a single motor, a number of LEDs and the radio transmitter. There will be at least two solar panels, each measuring voltages, one serving as a control variable and the other being used as part of the solar tilting experiment. The power output of these will also be implemented as a secondary objective. Humidity and external temperature must be read, with the option to include internal temperature of the payload needing to be considered. The input to the microcontroller from a GPS device is a necessary requirement

The microcontroller selected for this project is the Arduino Mega 2560 Rev 3, which is based around the Atmel Atmega 2560 chip. There were a number of reasons behind the choice of this particular chip and board. Firstly, the hardware requirements stipulate between six and ten analogue I/O ports. Considering this requirement against the other Arduino boards such as the Arduino Uno means that the Arduino Mega is the only available option, since all other boards have only six analogue ports, compared to the 16 available on the Arduino Mega. Other microcontroller chips were considered such as the PIC family of microcontrollers. In terms of complexity, these would require more work to get started, including the external circuitry and it was for this reason that these were discounted. Despite the lower price of the PIC family of microcontrollers, the extra time required for using the PIC chips and the higher confidence of reliability provided by the Arduino, meant that this was considered as the better option. This also allows focus to be placed on the hardware and software for the sensors, actuators and the design of the antennas.

4.3 HARDWARE AND SOFTWARE IMPLEMENTATION

The hardware and software for this project will be developed simultaneously, and will focus mainly around a single external board. This board will contain any external circuitry required for the operation of sensors and actuators. Throughout this section, each piece of the hardware will be designed individually and then, at end of this section, a final circuit design will be presented and will tie in with the final piece of software code which is included in the supporting material.

4.3.1 GPS TRANSCEIVER SYSTEM

The hardware for a GPS system is complex, involving the design of a circularly polarized antenna, and is a separate project in itself. Therefore, a pre-fabricated device was selected. The device selected was the Parallax PMB 688, which features 20 satellite tracking channels, meaning fast and constant GPS reception can be achieved. Perhaps the most useful aspect of this particular GPS receiver is that it has an on-board memory which will continue to transmit the last position, even if a signal is lost. This will provide an approximate location at least until signal is regained. The connection to the main microcontroller is simple, requiring only a single cable. This particular GPS unit provides National Maritime Electronics Association 0183 (NMEA) strings, which can be interpreted.

There are five GPS strings which are received by this device, and the uses of these are outlined in (El-Rabbany, 2002). The strings are received in the ASCII format and placed into a large buffer. Five different strings are received initially. The most useful of these is the GPGGA string, which is the only string which provides the altitude of the receiver. Therefore, the buffer is searched for the ASCII characters of "GPGGA". Of this string, the latitude, longitude, the time, the altitude and the quality of the GPS fix are useful. As such, the rest of the string is truncated. The remainder of the software for the operation of actuators and sensors is programmed around the manipulation of the NMEA strings. Provided below is a flow chart which shows how the GPS strings are extracted and formatted

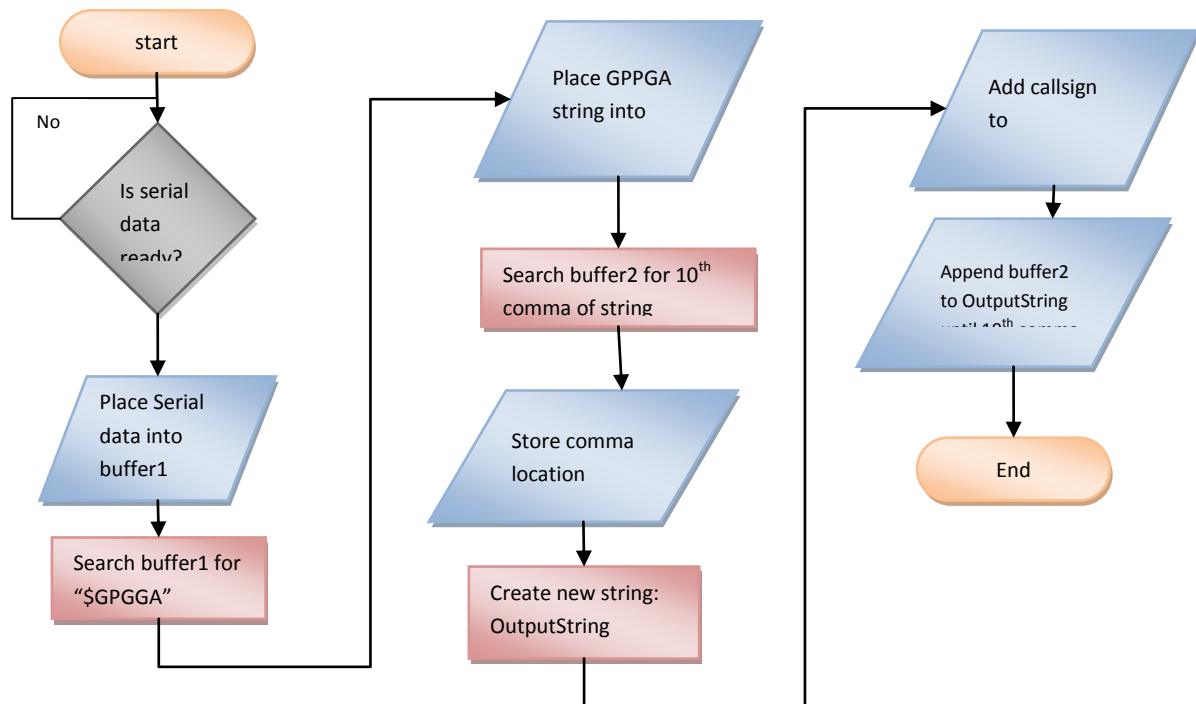


FIGURE 4: GPS STRING MANIPULATION: HIGH LEVEL FLOW CHART

4.3.2 SOLAR PANEL VOLTAGE MEASUREMENT

In order to determine how the altitude has an effect on the solar panel voltage, it is necessary to measure the voltage of the each panel. This can be done in quite a simple fashion. The Arduino can be used to measure the voltage using its internal reference. This will return a value between 0 and 1024, which can then be calibrated to provide an accurate measurement. This is simple to program in software, however, when implemented it was noted that there were large variations in the voltage. This was due to the resolution of the Arduino, and fluctuations in voltage. Initially, a 100 point average was used to counteract these variations, but this required more processing time and more memory. Therefore, an RC filter was used with a 0.5 second time constant as shown in the diagram below:

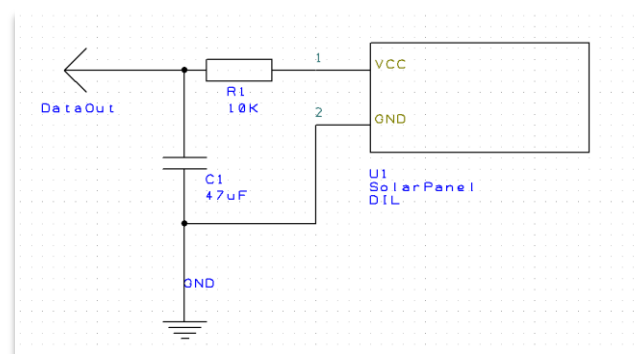


FIGURE 5: RC FILTER CIRCUIT CREATING 0.5s TIME CONSTANT

This provided a more stable voltage with less variations. To ensure further stability, a 20 point moving average filter remained in place. A comparison of methods is shown on the graph below.

This method proved successful in eliminating the voltage variations, and was implemented for the other voltage measurement such as those for the batteries.

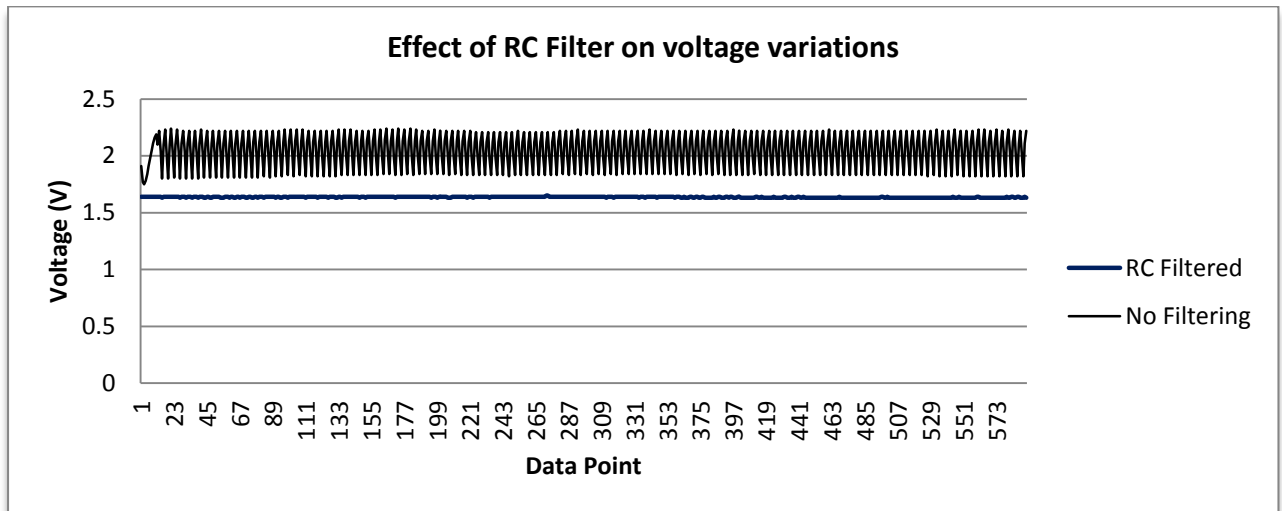


FIGURE 6: GRAPH SHOWING SMOOTHING EFFECT OF RC FILTER CIRCUIT ON VOLTAGE FROM SOLAR PANELS

It must be noted that the graph in Figure 6 shows the voltage from the solar panel being provided with varying light. The graph illustrates the way in which the filtering smoothes the values received, and does not have an effect on the voltage received.

4.3.3 STEPPER MOTOR CIRCUIT DESIGN

This is required to provide a tilting mechanism for the solar panel. Using a stepper motor instead of a regular DC motor provides a more accurate solution, allowing precise and exact steps through providing electrical pulses. A stepper motor offers more controllability and more precise movements. This makes it a good choice for an angular movement application. With this choice in mind, there are further choices for the motor configuration. There are two types of stepper motors: The permanent magnet motor and the variable reluctance stepper motor. The former offers a smaller stepping angle than the latter, but provides less torque. However, since it would be more useful in this application to have smaller step sizes than to have large amounts of torque, a permanent magnet motor was used (Shetty & Kolk, 2010).

Further to this, there are two types of permanent magnet motor, the uni-polar and the bi-polar stepper motor. A good choice can only be made once the differences are looked at, and this will also help in understanding how to program the motor. The main differences between these two types are the ways in which their coils are excited, and how the external pulses are applied to them. A typical uni-polar stepper motor will consist of six external wires. In the case of a bi-polar motor, there will typically be only four external wires. For a uni-polar motor, the two extra wires are typically connected to an external supply, with the other four supplying the pulses for the excitation of the different coil windings. Both types have advantages and disadvantages, but in this case, the uni-polar will be used since it requires less external circuitry over the bi-polar solution.

Since stepper motors require an exact sequence of pulses to begin moving and stay in motion, it is necessary to use a stepper motor driver chip otherwise rotation will be erratic at best. The chip being used in this case is the UCN5804B. This chip is readily available and provides three different

options for excitation of the motor. It is also able to provide the direction and can switch the supply voltage. This is useful since it can isolate the high current required for the driving of the motor from the central microprocessor.

Initially the motor was the Sanyo Denki 103H5210-0440. This motor was driven using a two-phase excitation mode, and as such the chip was configured as follows:

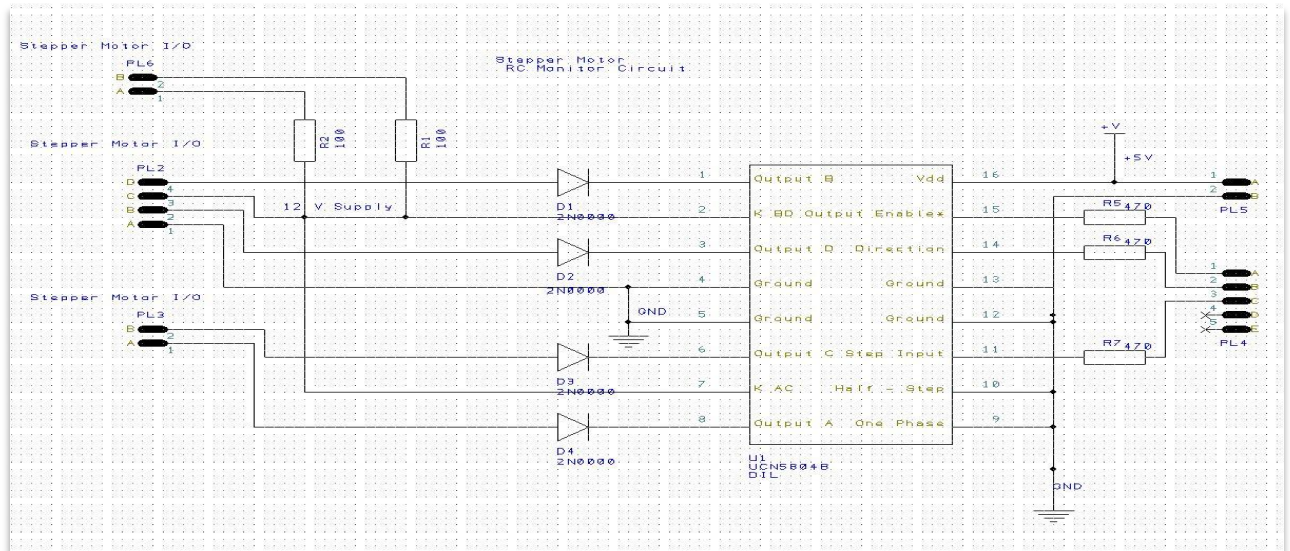


FIGURE 7: INITIAL STEPPER MOTOR WIRING DIAGRAM

As shown in the above diagram, the circuit was powered from a 12V supply and controlled by three pins from the micro-controller. The stepper motor is external to the circuit and shown by the I/O pins on the left. The two 100Ω resistors limit the current to the motor, allowing the motor to draw 0.12A per phase. Since the motor will run from a battery during the flight, this prevents rapid discharge of the battery, whilst still allowing sufficient torque for the motor to turn. The stepper motor supply is grounded through the common ground of the Arduino. Two phase drive format is achieved by forcing both pin 9 and 10 of the driver chip low. The 470Ω resistors provide a layer of isolation between the microcontroller pin and the stepper motor circuit.

Through the use of the stepper motor driver chip, it is possible to keep the software simple. All that is required is the relevant high and low signals to the three pins of the driver chip. In order to provide rotation, the 'Step Input' pin (pin 11) is driven high and then driven low after a certain delay. This is repeated a certain number of times. If this is repeated 200 times, the stepper motor makes a full 360 degree rotation. This is split into a separate function for efficient coding

To create the solar optimization angle system, the solar panel is mounted to the stepper motor, and the voltage is taken in at a range of angles. This is explained in the figure below:

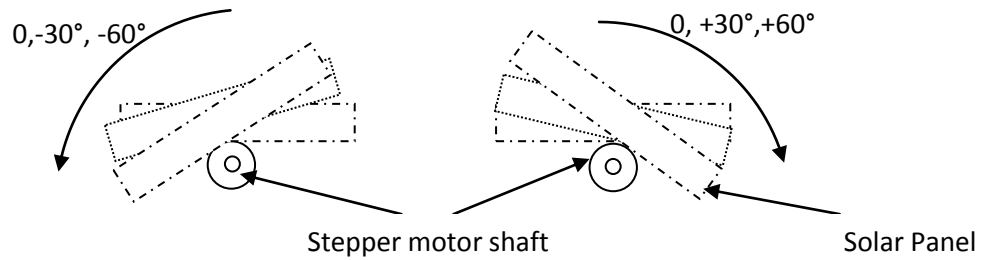


FIGURE 8: STEPPER MOTOR ROTATION

As shown in Figure 8, the stepper motor will rotate with the solar panel over a 120 degree arc divided into five separate points, with the origin being when the panel is horizontal. At each point, the voltage is read, and stored in an array. The position of the voltage in the array defines the angle at which that voltage was recorded. For example, the -60° position is the 1st element of the array, -30 is the 2nd element and so on until the 5th element is the voltage at the $+60$ degree position. The voltages are compared to find the maximum, and the panel is returned to the position where the maximum was recorded.

Further to the development of the payload, the Sanyo Denki 103H52110-0440 was replaced with a lighter, smaller stepper motor. This has a number of advantages. Firstly, being lighter allows the overall weight to be reduced, meaning that a faster rate of ascent can be achieved or less Helium gas can be used. The new motor is part of the Sanyo Denki 103H3 Series, and has smaller permanent magnets. This means that it is lighter, and requires less voltage to operate. However, it does mean that to create similar torque to the previous motor, the resistors limiting the current drawn from the battery need to be reduced. Reducing the resistor values from 100Ω to 47Ω doubles the current, allowing larger torque to be developed. At this point, the stepper driving mode was changed from two phase to half step mode. This mode provides smoother rotation, and requires less power, and whilst still maintaining similar torque characteristics to the two phase mode. The full code for the stepper motor rotation algorithm is included in the supporting material.

4.3.4 INTERNAL AND EXTERNAL TEMPERATURE SENSORS

External Temperature Sensor

The internal and external temperature sensors are a key part of the data collection system, and are also extremely useful during the testing phase of the project. The external temperature sensor being used in this case is a commonly used DHT-22 temperature and humidity sensor. This sensor was chosen for a number of reasons. The first reason is that it is a digital device. This means that valuable analogue pins are not needed, and are left free for solar panels and battery measurement systems. Secondly, this device incorporates an accurate temperature and humidity measurement system. The device can measure humidity to between 2% and 5% accuracy, and the temperature can be measured to within 0.5°C of the true value. It also has a wide temperature range (-40°C to $+150^\circ\text{C}$) when compared with other devices, such as the DHT11. Adding to this, the size of the device is relatively compact, and has a simple, single screw mounting system, allowing it to be fixed to the outside of any flat surface.

In order to use this device, a downloadable library is used. This library means that the temperature sensor is simple to code, through the use of two 'read' functions. The output of these two functions is stored in two floating point variables, which are then converted to character strings for transmission. The only external hardware that this device requires is a 10kΩ pull up resistor as shown in the diagram below:

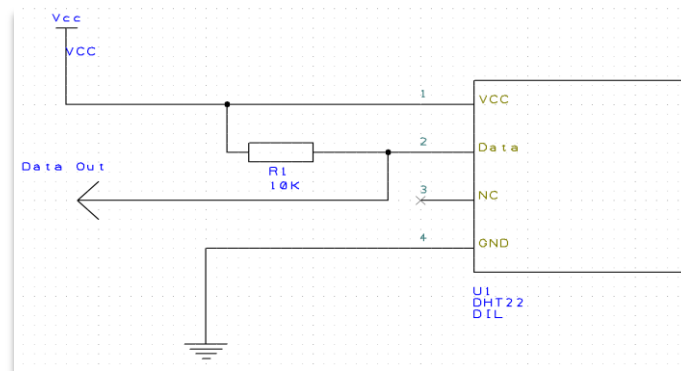


FIGURE 9: DHT22 EXTERNAL HARDWARE CONNECTIONS

This device, when configured in this way provides accurate readings of the temperature and humidity of the external environment, and has a large enough temperature range to measure the low temperatures of the upper atmosphere. Comparing this to analog temperature sensors, such as the LM35 or LM335, it does not require any calibration, and also can measure not only the temperature but also the humidity. It is also designed to be mounted away from a PCB or circuit board, meaning it is the ideal choice for this application.

Internal Temperature Sensor

The analogue devices mentioned previously, the LM35 and LM335 were considered for the internal temperature sensor. However, the most readily available device was a current controlled temperature sensor. The particular device used was the Analog Devices AD590 Temperature Transducer (Analog Devices, 2013). This device requires one external hardware component in the form of a resistor in series with the device. This provides an output voltage in which 0.1mV represents 1 degree Kelvin. The circuit diagram is shown below:

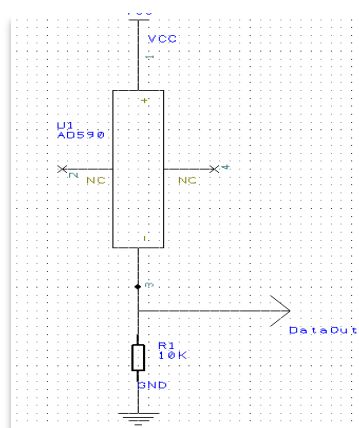


FIGURE 10: CIRCUIT DIAGRAM OF CURRENT CONTROLLED TEMPERATURE SENSOR

The 0.1mV/Kelvin change then needs to be adjusted in software to provide an accurate output due to the pull up resistors inside the Arduino. As this is an analog device, it is connected to an analog pin on the Arduino. This provides a value between 0 and 1024. This is then multiplied by 5 (the reference voltage) to provide a voltage. This then needs to be divided by the 10k Ω resistor to give a current value. It is at this point that the code needed modifying. Due to the internal pull-up resistors in the Arduino pins, the voltage and current values were erroneous. As such, rather than divide by 10k Ω , the division is over 9400 Ω , giving a more accurate result. This is then multiplied by 10^5 to provide a result in Kelvin. Finally the conversion from Kelvin to Celsius takes place, providing an accurate internal temperature sensor.

4.4 FINAL CIRCUIT DESIGN

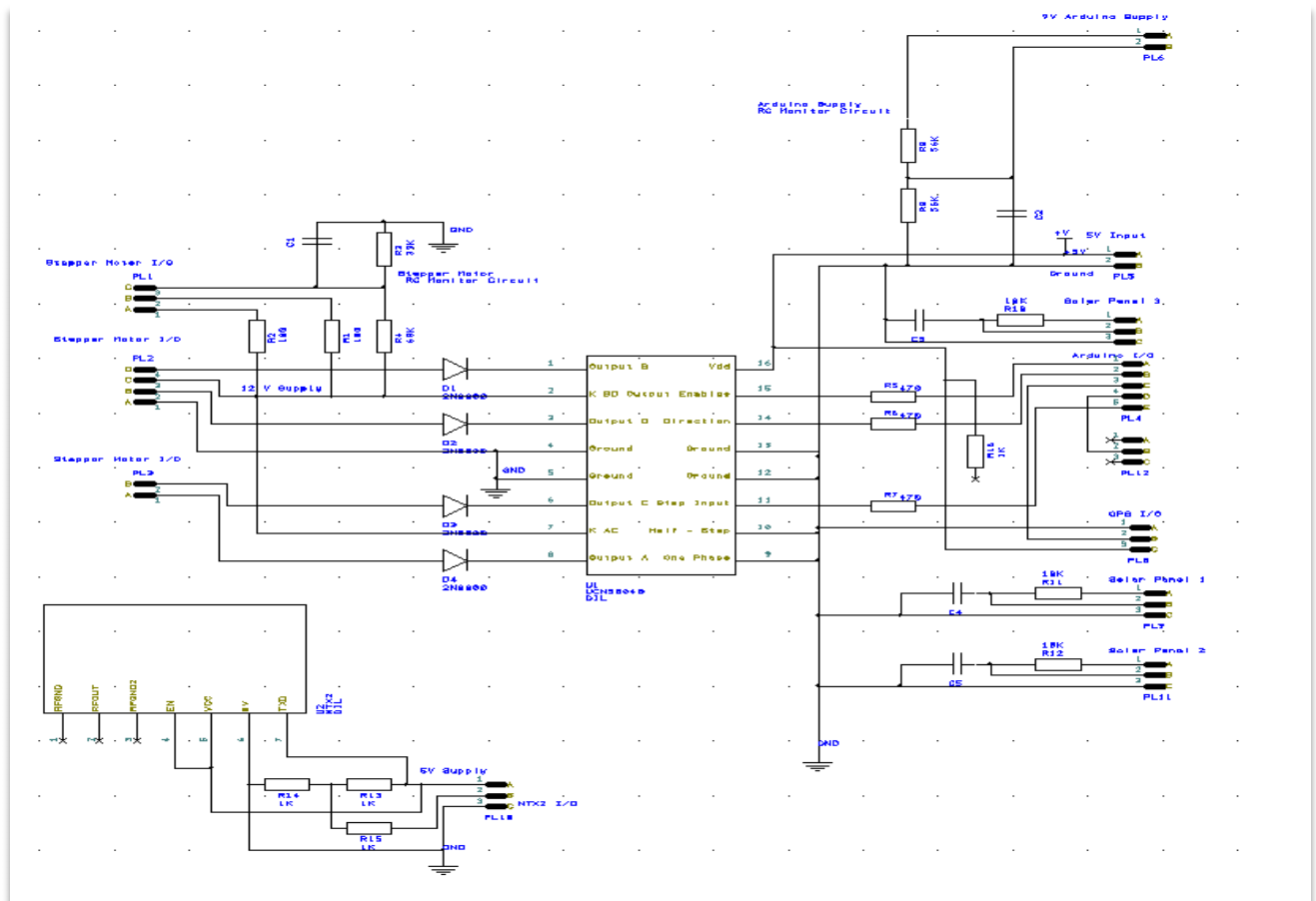


FIGURE 11: FINAL CIRCUIT DESIGN

The image above shows the final circuit design. On the right are the I/O connections to the various hardware components which have been detailed in the above sections. The bottom left shows the transmitter circuit which is developed at a later stage in this report. The full circuit will be powered from two 9V batteries. Appendix C shows the constructed circuit design.

4.5 PAYLOAD STRUCTURAL DESIGN

The structure of the payload is critical to the intact recovery of the instrumentation and recording equipment. There are a number of areas to consider, ensuring that the structure of the payload and the recovery systems are sufficient to enable retrieval of the system.

4.5.1 PAYLOAD MATERIALS

The material that the payload is constructed from impacts two areas of the design. The first element is the thermal insulation. This is due to the large variations in temperature during a high altitude flight. The second aspect is that of the structural integrity of the outer casing. This will provide a level of protection for the internal components, ensuring that the payload is recoverable at the end of the flight.

In order to meet the regulatory requirements for the flight in terms of the *payload*, it needs to be lower than a certain weight. Adding to this, by having a lesser weight, the balloon can be smaller requiring less gas. As such, a lightweight but strong material needs to be used. There are many plastic and fibre composites which would provide minimal weight and excellent protection, but lack the inherent insulating qualities that materials such as Polystyrene and Styrofoam have. Adding to this, plastics and materials such as acrylic are harder to cut and drill, whereas polystyrene is easily moldable into different shapes. Boxes are available widely for very little expense, and so these would provide an ideal starting point for testing, particularly in terms of how well these structures insulate. Further insulation may be required, and so foam pieces will be used to ensure that the payload is well insulated. Furthermore, a smaller size box will mean that any heat generated from the component operation will be kept in a smaller volume. Since weight is at a premium, to ensure the regulations are adhered to, a smaller box is also useful in this respect. As such a box of 19cm x 19cm x 13cm with 4.5cm thick polystyrene side walls was chosen.

4.5.2 PAYLOAD STRUCTURE

The location of the components and instrumentation is of critical importance to the safe and reliable operation of the payload. To ensure that the sensors, actuators and recording equipment are able to fit inside of the chosen payload size, model designs were carried out on Google Sketchup. This is shown illustrative purposes below. This design can be compared with the final structure shown in Appendix C

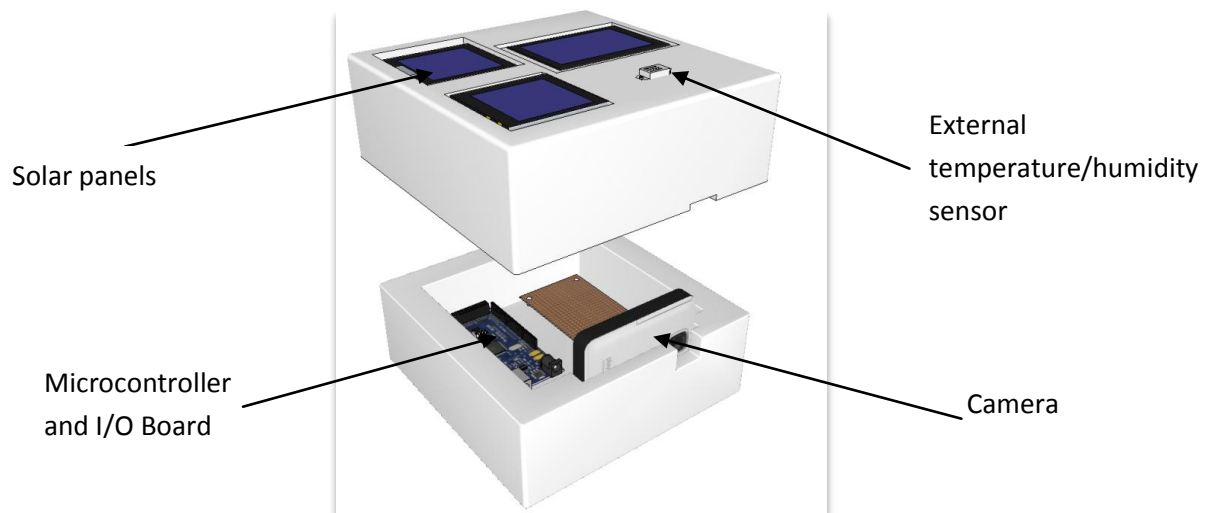


FIGURE 12: 3D DESIGN OF PLANNED PAYLOAD LAYOUT

4.5.3 BALLOON AND PARACHUTE CONSIDERATIONS

The actual calculations for the sizes of the balloon and the parachute are quite complex and begin to venture into mechanical engineering, and are perhaps beyond the scope of this report. A brief overview of the physical parameters which are considered is included however.

In terms of the parachute, it must be sufficient to slow the payload to a descent speed of between 3.5 and 5m/s. The actual decent speed will vary throughout the flight, since in the upper areas of the atmosphere there is less air per square meter than on the ground. The surface area of the parachute itself will also vary with the payload weight, in that a larger weight will require a larger parachute. This is due to the physical forces from gravity and air resistance. The UK High Altitude Society provides data on parachute sizes for different weights of payload, and this is included and referenced in

Appendix A. It is determined from this sizing chart that for a weight between 800 and 1000g, a parachute size of 33 inches or 0.83m diameter is required. This is constructed from rip-stop nylon. This material is chosen due to its high durability and its resistance to water degradation. The parachute design is a simple square with two eyelets in each corner

The altitude that is achievable is affected by three parameters: the payload weight, the balloon size and the volume of gas filling the balloon. There is also a trade-off involved between altitude and rate of ascent. The rate of ascent should be no less than 3m/s to prevent the balloon drifting due to the winds at high altitude; however, to achieve a higher rate of ascent, more gas is needed to provide more lift. This in turn leaves less room for the gas to expand meaning that the balloon will burst earlier. To compute the values of achievable altitude, ascent rate and gas volume requires a large number of calculations which are again slightly beyond the scope of this report. The target weight for the payload was defined to 800g, and this can be achieved using a 300g Balloon and 1.81m³ of helium gas.

4.6 INSTRUMENTATION AND PAYLOAD SUMMARY

In this section the each element of the payload's instrumentation and actuator control systems have been researched, designed and implemented both in software and in hardware. The payload itself has been modeled using 3D CAD tools and the relevant flight parameters in terms of the parachute and balloon size have been considered. The instrumentation will provide consistent and accurate data of the conditions of the atmosphere whilst other payload monitoring systems have been implemented to track the payload during the high altitude flight. However, these sensing systems and location tracking systems need to be transmitted from the payload to the ground using an RF transceiver system. This system is researched in detail and then designed and implemented in the next chapter of this report.

CHAPTER 5: TRANSCEIVER SYSTEM: RESEARCH, DESIGN AND IMPLEMENTATION

5.1 OVERVIEW AND DESIGN PROCEDURE

The transmission system is one of the most important and complex areas of the design. In terms of the basic hardware involved, the system requires a transmitter, a transmission antenna, a receiving antenna and a receiver. The receiver itself must also be connected to a user interface which will provide interpretation of the received transmission. The data from the various sensors on the payload must be modulated onto the carrier frequency of 434MHz as was defined in the regulatory requirements.

The procedure for designing the transceiver system will be to first design the hardware and software for modulation of the data. Following this, the receiving system will be looked at in terms of the conversion of the modulated data back into a textual format. Following this, the antennas for both the transmitting and the receiving systems will be designed. The antenna section will firstly look at the various antenna designs and their advantages and disadvantages for this particular application. Once a design is selected, it will be researched in terms of the necessary dimensions and relevant power transfer methods and this will then be used to define the dimensional requirements for that particular antenna. With the theoretical dimensions defined, the design will be simulated using two software packages: CST Microwave Studio and Sonnet High Frequency EM Simulator. The use of two packages will ensure that the results are consistent between each fabrication. Using these simulation packages, the dimensions and materials can be optimized for the necessary frequency. Both the ground and payload antennas are then fabricated, and the development of the fabricated designs is included in the next chapter, Chapter 6.

5.2 TRANSCIVER HARDWARE AND SOFTWARE

5.2.1 DATA ENCODING

There are a number of different ways of encoding data for transmission, but the most common method is to encode the textual data as ASCII (American Standard Code for Information Interchange) data. Source coding essentially ensures that when sending the information, the least number of bits is being used. ASCII code is primarily concerned with converting English text into encoded 1 and 0 pulses. (O'Reilly, 1984)

ASCII code contains all 26 letters of the English alphabet, in both upper and lower case, all numerals from 0 to 9, along with various punctuation and end of line characters. Each of these is represented by a unique 7 bit binary code. Modifications on ASCII provide 8 bit binary coding, allowing more characters, but the most basic form is 7 bit ASCII. Further modifications on ASCII include systems which use fewer bits to transmit more common characters, however these are slightly more complex to encode in software and more likely to introduce errors due to the variable length of the message data. Reception of this data is also more complicated, since each character may not have the same number of binary bits. As such, to ensure that data is correctly interpreted, and to minimize the complexity of the software encoding, basic 7 bit ASCII will be used to encode the data. (O'Reilly, 1984). The data also requires a checksum to be appended to the string. A standard checksum must be used so that receivers across the country can interpret the data and upload the strings to the mapping software.

The technique for encoding the software is relatively straightforward. Firstly all the data must be converted into string format. This can be done using

Following on from this, it is necessary to consider how to transmit the encoded data. There are a number of methods which can be used, and these are considered in the next chapter

5.2.2 MODULATION

Since the regulatory requirements set out in Chapter 3 specify that the transmission must be within a certain frequency band, and at a certain power level, it is necessary to modulate the data onto a carrier frequency. The band chosen was the 434MHz band due to its limited interference potential, and the availability of transmitter systems. The UK High Altitude Society recommends that for this frequency, the NTX2 Module from Radiometrix is used. This module satisfies the power and frequency requirements for this frequency band.

The use of the ASCII coding technique, combined with the analog FM transmitter, means that there are only a few methods for transmitting the data. Modulation requires manipulation of at least one characteristic of a carrier signal. Due to the fact that the transmitter is an FM transmitter, the modulation is having an impact on the frequency of the carrier signal. The combination of the digital encoding, the analog transmission medium and the FM transmitter limits the forms of modulation.

One of the most common methods of transmitting digital data with Frequency Modulation is to use Binary Frequency Shift Keying (BFSK). This is a form of Frequency Modulation in which the two binary values are represented by two separate frequencies. The standard form of BFSK is to have a carrier frequency, and two different frequencies offset from the carrier frequency, such that one can be used for a binary '1' and the other can be used for a binary '0'. (Stallings, 2009)

One of the simplest forms of BFSK is to use Radio Teletype (RTTY). This system is now largely obsolete and is now only commonly used in Amateur radio bands. RTTY consists of only two frequencies: a 'mark' frequency and a 'space' frequency. The mark and space frequencies represent the binary 1 and the binary 0 respectively. Using this method it is possible to transmit binary data at approximately 50baud. (Ford, 2008). A typical frequency shift for the 'space' frequency only needs to be within 100 to 800Hz of the 'mark' frequency.

It is also important to consider a new alternative method for transmitting, which was designed for amateur use. This method is known as PSK – 31. PSK is phase shift keying, and it involves the manipulation of the phase, rather than the frequency. This method combines some of the good qualities of RTTY transmission with those of a variable length encoding system such as Morse code. It is however, relatively untested in high altitude balloons and since data transmission is a crucial aspect of the system, a more commonly used method may be more suitable.

RTTY will be the chosen method for this particular task. This is for a number of reasons: Firstly, the technology has been in use for over 50 years, and was first used for long range military communications. This shows that it will work over the long distances involved in this project. Adding to this, the transceiver system can be relatively inexpensive, since RTTY can be received using any Single Side Band receiver (Ford, 2008). With this choice in mind, it is necessary to now consider how to use the NTX2 to provide the two mark and space frequencies.

5.2.3 NTX2 OPERATION

The Radiometrix NTX2 transmitter is a low power RF transmitter, which operates at 434.65MHz. It can transmit up to 10mW, meaning a reasonably slow data rate of approximately 10kbps. (Radiometrix, 2012) This means that it satisfies the regulatory requirements for the transmitter frequency and power rating. The NTX2 contains a voltage control crystal oscillator (VCXO), power amplifiers and various band-pass and low pass filters. This provides an output of 434.65MHz when enabled and provided with a constant voltage between 2.9V and 15V on the Vcc Pin. In order to use the transmitter to send data, the 'TXD' pin voltage needs to be varied. Variations on the TXD pin need to be between 0 and 3V, which will provide up to a 5kHz frequency variation.

Data can be transmitted in many ways, but using RTTY means that it is necessary to have two different frequencies: one of these can represent a '0' databit, the other can represent a '1' databit.

From looking at the datasheet, it is clear that the transmitter needs to be provided with two voltage sources. One source provides the power, whilst the other source can be variable, on the TXD pin. The voltage input at the TXD pin needs to be between 0 and 3V. Using a voltage divider, the 5V Vcc from a digital pin can be brought down to provide a bias voltage between 0 and 3V. Using the fact that a 3V shift on the TXD pin will provide a 5kHz shift, it is simple to calculate that for a 1Hz frequency shift, a voltage of 0.0006V needs to be applied to the TXD pin. A reasonable variation in the frequency would be 650Hz, meaning that the voltage on the TXD pin needs to be varied by 0.39Volts. This variation can be provided by a third resistor connected in series with a 5V digital pin.

The external hardware for the transmitter circuit is shown below:

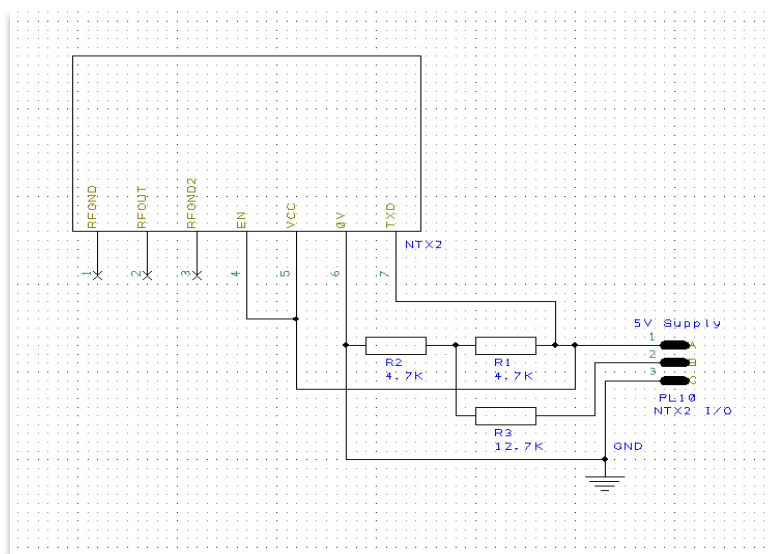


FIGURE 13: NTX2 EXTERNAL HARDWARE CONNECTIONS

The 5V supply is provided by the Arduino, and is used as part of the biasing for the TXD pin. A potential divider network formed by R1 and R2 is designed such that the voltage on the TXD pin is biased to 2.5V:

$$V_{TXD} = \frac{R2}{R1 + R2} \cdot V_{in}$$

R1 and R2 are typically chosen so that the sum of the two resistances is approximately 10k Ω , lowering the power dissipation through the potential divider:

$$2.5 = \frac{R2}{10000} \cdot 5$$

Therefore, R2 will be:

$$R2 = \frac{2.5 \times 10000}{5} = 5000\Omega$$

This means both R1 and R2 need to be chosen to be approximately 5k Ω . Both resistors used will therefore be the widely available 4.7k Ω resistors. This potential divider circuit provides the biasing of the TXD pin. This means that with no voltage applied to the TXD input, the TXD voltage will be 2.5V. This correlates to a transmission frequency of 434 654 166 Hz for a '0' pulse. To achieve single sideband frequency modulation, a second frequency is used. Above, it was calculated that a 0.39 voltage shift would be sufficient to provide a 650Hz frequency variation. This means that the voltage for a high digital pulse will be 2.89V. Again the potential divider calculation can be used, with the aim of calculating a new effective value for R1.

$$V_{TXD} = \frac{R2}{R1 + R2} \cdot V_{in}$$

$$2.89 = \frac{4.7k\Omega}{R1_{eff} + 4.7k\Omega} \cdot 5$$

$$R1_{eff} = 3.431k\Omega$$

This is the effective value that R1 needs to equal however, as shown in Figure 13, R1_{eff} is formed of R3 and R1 in parallel. Since R1 was chosen to be 5k Ω , R3 needs to be chosen in relation to this:

$$\frac{1}{3.431k\Omega} = \frac{1}{4.7k\Omega} + \frac{1}{R3}$$

$$R3 = 12.7k\Omega$$

Choosing these values will provide two voltages of 2.5V and 2.89V, which correlate to two separate frequencies of 434 654 166 Hz and 434 654 816 Hz for the low and high pulses respectively. In doing so, this has created a method of transmitting textual strings as digital pulses, which are then transmitted as two separate frequencies. From this, the transmitter is connected to a transmission antenna. The design of this antenna is detailed in Chapter 5, Section 3. Once the signal is transmitted, it must be received on the ground through a second antenna. This antenna is also designed later. This antenna is connected to a receiver system, which must be capable of receiving single side band transmission. There are different methods to achieve this, which are detailed in the next section.

5.2.4 RECEIVER SYSTEM

The receiver system consists of three main components. The first of these is the ground antenna. This is designed in the next chapter. The ground antenna is connected to a receiver capable of receiving the Single Side Band RTTY.

There are two options for the receiver itself. The first is to use an FM radio, capable of receiving Single Side Band. Typically these radios are the preferred option, however they are expensive. Commonly used devices include the Yaesu FT817 or the ICOM IC-7000. Both of these cost in excess of £100, and as such are outside the budget for this project.

Alternatively, it is now possible to use a device called an SDR Dongle. These usually have a large band of operation, and as such can be quite noisy. However, when combined with two well made antennas, the signal to noise ratio is generally high enough to ensure that the signal can be received. A further disadvantage of these dongles is that the connector to an antenna is usually a PAL type connector. These have a passband of 0 to 100MHz, and as such have quite poor matching outside of this band. To solve this problem the PAL connector can be changed to a standard female SMA connector.

Despite the disadvantages of the dongle, it is the most cost effective option for receiving the data from the balloon. There are many devices available, each based on the RTL 2832U chipset. The particular device used in this case is similar to all other available devices.

In order to receive the signal (but not the data) from the balloon, the device requires a Software Defined Radio application. The application chosen in this case is called SDR Sharp (SDR #). This software has digital signal processing built into the user environment. Below is the view of the frequency spike at 434MHz:

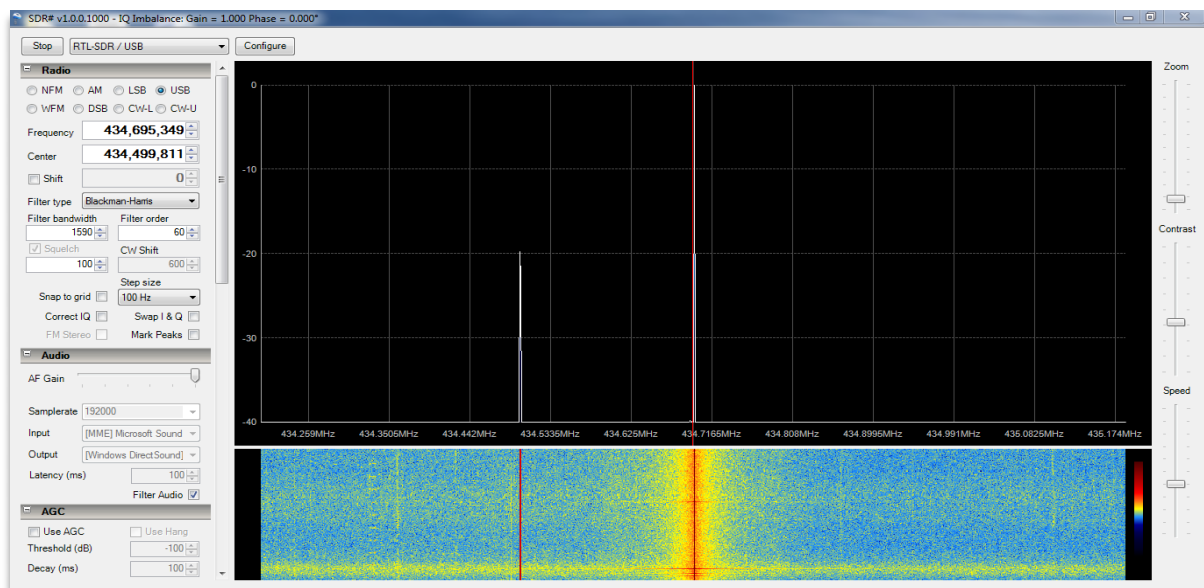


FIGURE 14: OUTPUT OF SDR# SOFTWARE

The above diagram shows both the 434MHz signal (the larger amplitude) and a DC offset (smaller amplitude). The DC offset does not have an effect on the received signal. What is important to note is that the above image shows the signal using the 'range offset' function in the SDR# user interface. In reality, from -40dBm downwards is the instrumentation noise floor. This is quite high for any system but as a reference, the best receivers have noise floors of around -115dBm (Carr, 2000). Of course this low level of noise floor is unachievable with this receiver; however it may be possible to improve the noise floor of this device into more acceptable levels.

The SDR dongle and the software are the initial stages of decoding the data. Looking closely at the frequency spike shows that there are actually two frequencies, which are the high and low digital bit stream. These are separated by the shift introduced by the voltage divider network that was mentioned previously. Although it is not possible to show both frequencies, it is possible to view the digital bits in the 'waterfall' below the frequency spectrum display:

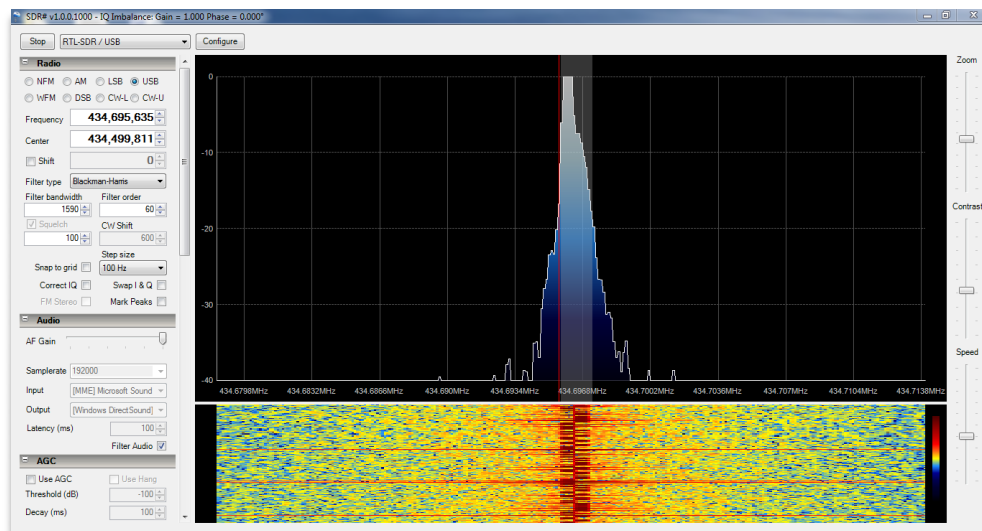


FIGURE 15: CLOSE VIEW OF HIGH AND LOW FREQUENCY PULSES

This is the received frequency spectrum, which now needs to be used to decode the bit stream into actual ASCII characters. To do this, a second piece of software is used, DL-FLDIGI. This software can decode a number of different types of encoded data, however for this purpose it is set to decode RTTY. The software requests entry of a 'carrier shift', which is the spacing between the two frequencies. In this case, it is set to 650Hz. The received signal has to then be inputted into the DL-FLDIGI decoding software, and this is done internally through the PC's internal audio jack. The output on DL-FLDIGI will be the original decoded data, as shown in the figure below:

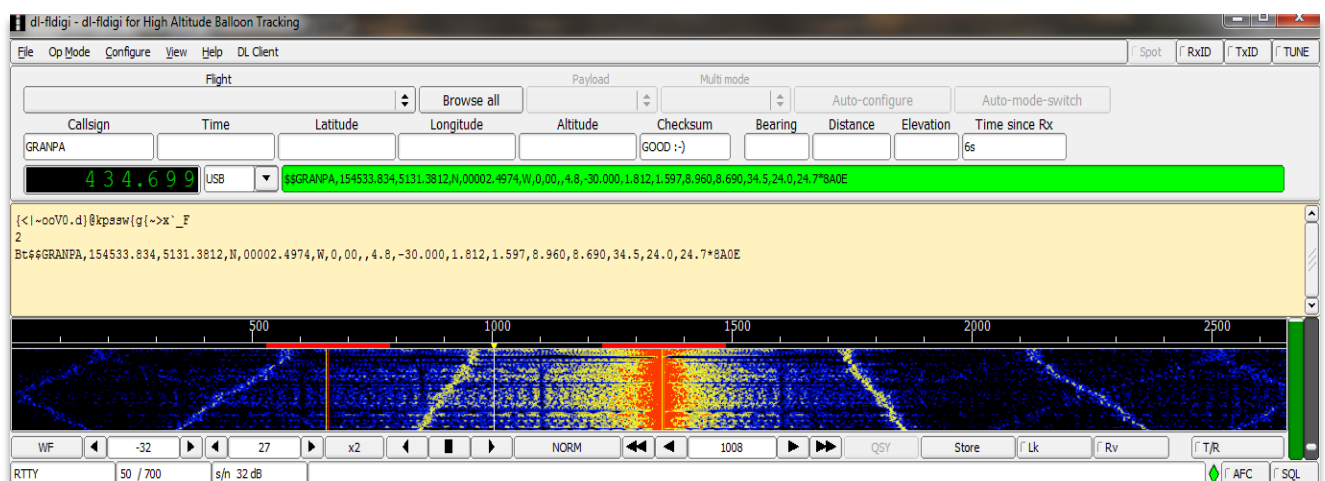


FIGURE 16: IMAGE OF DL-FLDIGI DURING DATA DECODING PROCESS

The decoded output strings can then be plotted on a map in order to track the balloon flight. The next stage is to design the antenna which will enable transmission of the encoded data to the ground during the flight.

5.2.5 MAP TRACKING SYSTEM

Before moving on to the design of the antennas, it is necessary to briefly mention the map tracking software. The software is an online software in which the strings from DL-FLDIGI are parsed, uploaded and stored in a server. This allows the data to be retrieved post-launch, and means that there is no need to record the data manually. In order to find the position of the balloon, the relevant data is uploaded to a map which is accessible from www.spacenear.us. There is a network of tracking stations across the UK which can receive the data on their own versions of DL-FLDIGI throughout the flight, and the data is parsed using the flight-documentation which is set up before the flight. Those strings which pass the checksum are uploaded to the server and the position is overlaid onto the map.

5.3 PAYLOAD ANTENNA DESIGN RESEARCH

5.3.1 DESIGN ALTERNATIVES

There are a large number of different antenna designs which can be considered, and as such, it is necessary to consider the design choices before beginning to simulate a particular type of antenna. The choice of antenna is based upon a number of factors, including the radiation pattern, the polarization, and the geometrical size of the antenna itself. The first aspect to consider is the polarization of the antenna. A linearly polarized antenna has a radiating wave in only one direction (E_θ) such as a dipole antenna. (Huang & Boyle, 2008) However, due to the predicted rotating nature of the balloon, it would be more advisable to use a circular polarized antenna, which would mean the wave would radiate in all directions (E_θ and E_ϕ). It is worth noting that circularly polarized antennas are commonly used for satellite communications due to the Faraday Rotation in the ionosphere. Faraday rotation is caused by the high electron content in regions of the upper atmosphere, and causes the plane of the polarization of the wave to be rotated over the propagation region. The wave can be rotated by up to 150 degrees (depending on conditions of the atmosphere and the frequency of the signal) (Maini & Agrawal, 2011)

The requirement of circular polarization eliminates a significant number of antennas, including the monopole and the linear polarized patch antenna. With the circular polarization requirement in mind, two antennas can be considered for this application: a circularly polarized patch antenna, and the helical antenna, which is, by design circularly polarized.

Another important aspect to consider, based mostly upon the regulations outlined in chapter 3, is the weight of the antenna. The helical antenna, at the frequencies of this project, is quite large. A very brief calculation using internet based calculators¹ shows that the helical antenna would be over 30cm in length. If this is made from copper wire, combined with a ground plane as per the standard helical design, the weight may become quite a significant concern. Comparing this with a circularly

¹ <http://jcoppens.com/ant/helix/calc.en.php>

polarized patch antenna shows that the patch antenna is favorable due to its weight alone. Adding to this, the fact that the patch antenna is planar, and is therefore low profile means that the safety risk upon landing is reduced. These factors make the patch antenna the preferred choice for the payload antenna. There are limitations to the patch antenna, in that it can have a smaller bandwidth and lower gain than conventional microwave antennas however there are ways to improve these parameters, and like any antenna design, it is tradeoff between requirements and the features of the antenna. (Garg, et al., 2001).

5.3.2 BASIC DESIGN

With this choice in mind, it is worth briefly looking at how the design of the patch antenna allows it to radiate. This will be considered first from a linearly polarized point of view, and then from a circularly. (Garg, et al., 2001) describes this clearly starting from the basis that any microstrip line will radiate, and while in microstrip circuitry, this may be undesirable, it is this fact that is employed and exploited to provide the radiation. A patch antenna is generally a flat metallic ground plane, with a substrate of a certain dielectric constant ϵ_r with the patch shape etched into another metallic surface on top of this. Typical feeding methods are through transmission lines or through a coaxial feed through the layers of the patch. A coaxial fed patch antenna is depicted in Figure 17.

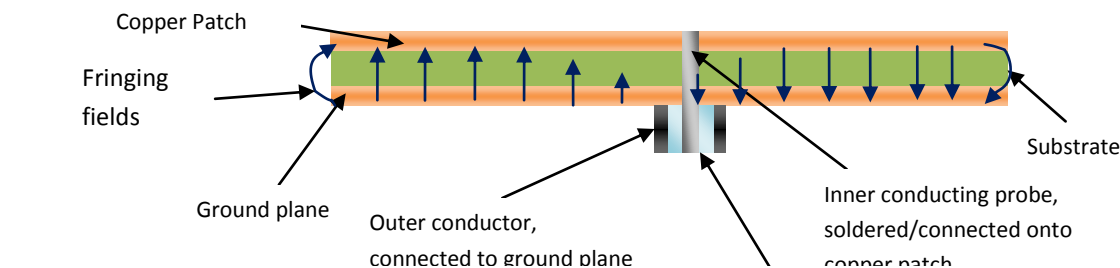


FIGURE 17: DEPICTION OF A COAXIAL FED PATCH

The electric field is underneath the patch. At the centre of the patch the electric field is zero, while at either edge, there is a maximum. This is shown in the figure above. The electric field continues at the edge of the patch, and these are known as the fringing fields. Put in simple terms, it is these fringing fields which cause the patch to radiate (Orban & Moernaut, 2005).

The dimensions of the patch antenna are determined by the wavelength in question and the substrate being used. For a square patch antenna, the resonant length (L) is typically defined as:

$$L = 0.5 \frac{\lambda_0}{\epsilon_r}$$

EQUATION 1: THEORETICAL PATCH ANTENNA RESONANT LENGTH

Other factors such as the ground plane, the thickness of the patch material and the width of the patch itself also influence the resonant length. (Orban & Moernaut, 2005) These should also be considered when calculating the dimensions of the patch.

5.3.3 FEEDING TECHNIQUES

Aside from the dimensions of the patch, there are other aspects to consider when designing the patch antenna. The feeding technique is one of the most important areas to consider, as it impacts the power transfer between the source and the patch. There are a number of feeding techniques which can be considered and the key factor when choosing one should be the minimization of undesired radiation. Unpredicted and undesired radiation from the feed can impact the radiation pattern of the antenna itself and will also have an effect on the overall efficiency. (Garg, et al., 2001) The feeding technique also impacts on the ease of manufacture and the how durable the patch is. (Waterhouse, 2003)

A coaxial fed patch antenna is one of the most basic forms of feeding, but it has limitations over other techniques. The inner conductor of an SMA connector is fed through the ground plane and substrate and attached to the metal conductor of the patch. The location at which the feed is placed impacts on the impedance match of the feed. Since the electrical field is minimum in the centre of the patch and at a maximum or minimum (depending on the phase) at the edges, it is clear that there will be maximum impedance at the edges and minimum or zero impedance in the centre. Typically, the impedance at the edges can be around 200Ω , therefore, there is point between the centre and the edges at which the impedance of the feed can be matched with the impedance of the patch (Orban & Moernaut, 2005). By ensuring that there is a good impedance match, the antenna gain and efficiency can be maximized.

An alternative to this method would be to edge feed the patch. This uses a piece of microstrip transmission line which is etched onto the same substrate as the patch, and is matched accordingly. This method has more advantages over the coaxial feed method, but involves a similar method of matching as the coaxial feed method. The feed should be inserted at a point along one of the radiating edges, and the location of which affects the impedance match. This type of feeding technique typically creates an antenna with a narrow bandwidth, but also can cause spurious radiation from the feed itself (Waterhouse, 2003). Circular polarization can be achieved using a simple power divider circuit such as a Wilkinson power divider or quadrature hybrid. (methods for circularly polarizing the patch are covered in more detail later).

Further methods of feeding include proximity coupling techniques. (Waterhouse, 2003) provides a very useful analysis of this kind of method. Using this method requires careful design, and is more complex than the aforementioned techniques. Figure 18 provides a pictorial demonstration of this technique. First the microstrip antenna is etched on top of a single piece of substrate as per the standard design. A feed is then located on a second, grounded substrate. There is no ground plane between the two substrates, and it is this fact that allows the power to radiate to the patch itself, without a direct connection between the two conductive elements. The primary advantage to this method is that the coupling to the patch is capacitive by its design, whereas previous methods are inductive. This means that a wider bandwidth design is created, but this is done at the expense of high spurious radiation. Fabrication can also be more difficult as air gaps have an impact on the effectiveness of the power transfer between the feed and the patch.

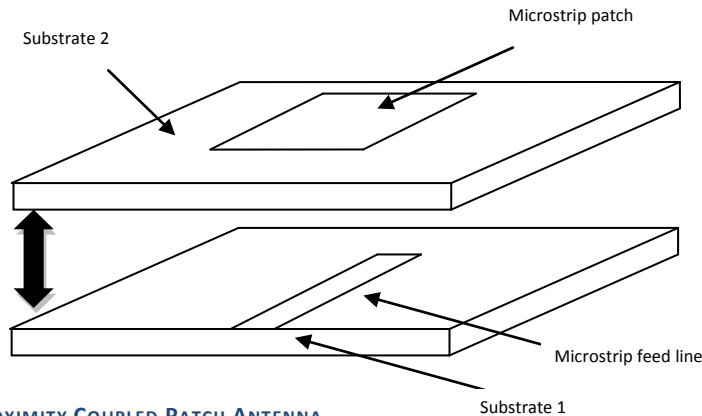


FIGURE 18: PROXIMITY COUPLED PATCH ANTENNA

These are a few of the most commonly used techniques which can be used to feed a patch antenna. There is no ideal choice which provides the best attributes in all cases, so a compromise is made. The choice for this design will be the coaxial feeding technique. This is mostly due to its simplicity in design and the ease at which the impedance can be matched between the probe and the patch. It is chosen over the transmission line feed since this method creates more spurious radiation which is undesirable. Edge feeding will also take more substrate area, meaning the potential for further weight addition. Weight is also a factor in eliminating the proximity coupling technique for this use, whilst the properties of the technique also mean that it is less ideal for this use. Coaxially feeding the antenna means that the ways of achieving circular polarization are based more on the patch shape itself rather than on feeding the patch differently. The methods for achieving the circular polarization are analysed in the next chapter.

5.3.4 CIRCULAR POLARIZATION CONSIDERATIONS AND TECHNIQUES.

The antenna coaxial feeding method limits the possibilities for achieving circular polarization. Circular polarization requires that the electric field varies in two directions equally, both horizontally and vertically. Each direction should have an equal magnitude in each orthogonal plane and a 90 degree phase shift between the electric fields. This is defined as two separate modes: TM_{01} and TM_{10} . (Lee & Luk, 2010) defines circular polarization as:

$$\vec{E} = (A\hat{u}_1 \pm jB\hat{u}_2)E_0$$

EQUATION 2: CIRCULAR POLARIZATION DEFINITION

Where u_1 and u_2 are two perpendicular unit vectors of an electromagnetic wave travelling in the perpendicular u_3 direction, with A and B being positive, equal, real numbers. (if A and B are not equal, elliptical polarization will be achieved). It is clear that from the above equation that the resultant electric field will have one of two directions based upon the positive or negative sign. A positive sign in the above equation results in Left Hand Circular Polarization (LHCP) whilst a negative sign will generate Right Hand Circular Polarization (RHCP). In the ideal case, an antenna should have only RHCP or LHCP, but in reality there may be a component in the opposite rotation, and this is known as cross polarization. In the ideal case the antenna should have co-polarization, in which there is only one sense of rotation. (Lee & Luk, 2010)

The other consideration for the patch antenna is that of the axial ratio. The axial ratio is essentially a measure of how well circular polarization has been achieved. Referring to Equation 2 on the previous page, the axial ratio would be defined as the ratio of signal B to signal A

$$\text{Axial Ratio} = \frac{A}{B} = \frac{E_{\theta}}{E_{\varphi}}$$

EQUATION 3: AXIAL RATION (AR) DEFINED

Most literature refers to the fields in the E_{θ} and E_{φ} directions, and this is a better way of expressing the axial ratio. Therefore, it can be seen that a linear polarized antenna would have an infinite axial ratio, whilst the aim for circular polarization is to have an axial ratio of 1, or 0dB. For a particular antenna, the axial ratio will be 0dB (or close to 0dB) at only one frequency, f_0 . The axial ratio will be greater or less than 0dB over a range of frequencies above and below f_0 . This gives rise to a bandwidth over which the axial ratio stays below a certain value and this is known as the axial ratio bandwidth.

In order to achieve circular polarization in a patch antenna, there are a number of standard methods. One technique has already been mentioned briefly. The use of power dividers combined with a transmission line feed was dismissed due to the potential for spurious radiation and the extra size that it introduces. Any method should not only satisfy the requirements for circular polarization, but also provide better characteristics than a modified transmission line feed.

To avoid changing the shape of the patch antenna, it is possible to use a dual probe fed solution. Two feed points located on two perpendicular edges, combined with a 90 degree delay in phase in one signal line will create circular polarization. However, this method can be considered as slightly wasteful and will take extra design time, since both feed points must be matched. Each feed will also cause some power to be lost, decreasing efficiency.

The losses mentioned can be reduced by using single feed technique, as outlined in (Sharm & Gupta, 1983) is to locate the feed along the diagonal of a square or nearly square patch. An alternative method to generate circular polarization is to alter the shape of the patch to generate the requirements of circular polarization.

A common method is to truncate two of the four corners of the patch antenna. This method is perhaps one of the simplest for providing circular polarization, and does not require slots and only uses a single feed. Care is required however, as this kind of design does tend to cause narrowband circular polarization. However, for its simplicity, this drawback may be acceptable (Imbriale, et al., 2012). Other techniques include circular designs with multiple feed points, or elliptical shaped patches or slots cut into square patches. For this design, two methods will initially be considered. The first method being the square patch, fed along the diagonal and the other being the truncated edge patch.

5.3.5 RESEARCH SUMMARY

The requirements set out in chapter 3 contributed to the selection of a particular antenna design, especially when considering the weight and the size of the antenna. By selecting a patch antenna, it is possible to design a relatively compact and light-weight antenna, with good characteristics. The

basic design equations provide a starting point from which simulations can be carried out. The feeding techniques are important when considering the design of the patch antenna, with the coaxial fed patch being chosen for its fabrication simplicity, with reasonably efficient transfer of power to the patch. The application of the antenna in the unstable environment of the atmosphere, means that circular polarization is a key requirement to ensure consistent operation. The methods for achieving this have been considered, and two techniques will be tested through simulations.

There are a large number of other factors which can be used to describe the antenna performance, and these will be expanded upon when analysing the antenna simulation results themselves. Some of these include the main lobe direction, beamwidth, front to back ratio, directivity, efficiency, gain and the aforementioned polarization. (Zhi & Chia, 2006).

5.4 PAYLOAD ANTENNA DESIGN AND SIMULATION

5.4.1 INITIAL DIMENSIONS

The initial design of a patch antenna stems from an equation for the resonant length of the patch antenna. The frequency at which the patch antenna resonates is determined by its length. Equation 1 of this report provides the basic relationship, however a closer estimate can be achieved by using the equations found in (Bahl, et al., 1982). The resonant frequency is related to the length and the dielectric constant of the substrate. In order to calculate the dimensions, first the substrate must be defined. In this case, the substrate chosen was FR4, which is a “fiberglass-epoxy based composite” (Lee, 2004). It is readily available and commonly used, and as such, has a low price. The problems with FR4 are that it is lossy compared to other substrate materials, and is often not of high quality. As such, the dielectric constant tends to vary between manufacturers and can change by up to 5%. It is generally considered to be 4.3, but has been at measured at 4.7 from some manufacturers. (Lee, 2004). At relatively low frequencies, variations can be tolerated and factored into the design. The standard thickness being used is 1.575mm

$$f_r = \frac{c}{2(L + 2\Delta l)\sqrt{\epsilon_e}}$$

EQUATION 4: RESONANT FREQUENCY PREDICTION

Where Δl is the effective increase of length of the patch due to the fringing effects through the substrate and is determined by the relationship between the height, width and substrate as follows:

$$\Delta l = 0.142h \frac{\epsilon_e + 0.3}{\epsilon_e - 0.258} \cdot \frac{\frac{W}{h} + 0.264}{\frac{W}{h} + 0.8}$$

EQUATION 5: EFFECTIVE INCREASE OF LENGTH

Again, due to the fringing effects of the patch, the dielectric properties alter slightly, giving rise to an effective dielectric constant which is determined by the following equation:

$$\varepsilon_e = \frac{(\varepsilon_r + 1)}{2} + \frac{(\varepsilon_r - 1)}{2} \cdot \frac{1}{\left(1 + 12h/W\right)^{1/2}}$$

EQUATION 6: EFFECTIVE DIELECTRIC CONSTANT

Re-arranging Equation 4 provides an accurate estimation of what the length of the patch should be, based upon the resonant frequency required.

$$L = \frac{c}{2f_r\sqrt{\varepsilon_e}} - 2\Delta l$$

EQUATION 7: ESTIMATION OF RESONANT LENGTH OF PATCH

Finally, the equation to estimate the width of patch is given by (Bahl, et al., 1982) as:

$$W = \frac{\lambda_0}{2} \left(\frac{\varepsilon_r + 1}{2} \right)^{-1/2}$$

EQUATION 8: WIDTH OF PATCH ANTENNA

Using the above set of equations provides the following dimensions and parameters:

$$W = \frac{\lambda_0}{2} \left(\frac{\varepsilon_r + 1}{2} \right)^{-1/2} = \frac{0.6912}{2} \cdot \left(\frac{4.3 + 1}{2} \right)^{-1/2} = \mathbf{0.2123m}$$

$$\varepsilon_e = \frac{(\varepsilon_r + 1)}{2} + \frac{(\varepsilon_r - 1)}{2} \cdot \frac{1}{\left(1 + 12h/W\right)^{1/2}} = \mathbf{4.2311}$$

$$\Delta l = 0.142h \frac{\varepsilon_e + 0.3}{\varepsilon_e - 0.258} \cdot \frac{W/h + 0.264}{W/h + 0.8} = \mathbf{2.5405 \times 10^{-4}m}$$

$$L = \frac{c}{2f_r\sqrt{\varepsilon_e}} - 2\Delta l = \mathbf{0.16751m}$$

The above equations provide the dimensions for the square patch antenna, and these can then form the basis for the dimensions of the truncated edge patch antenna. The location of the feed point is crucial for impedance matching, efficiency and the return loss results (Yong & Bernhard, 2009).

When considering the location of the feed point, it is necessary to look at how the feed will arrive at the patch, and then match the location at which the patch is fed to the impedance of the feed. A coaxial fed patch is typically fed using an SMA connector which commonly has a characteristic impedance of 50Ω. This means that the location should be chosen to match this impedance.

An approximate position can be determined through two equations in (Gustrau, 2012):

$$y_f = \frac{W}{2} \text{ and } x_f = \frac{L}{2\sqrt{\varepsilon_e}}$$

EQUATION 9: LOCATION OF COAXIAL FEED POINT FORMULAE

In order to achieve circular polarization with a square patch, the research earlier in this report has shown that the feed point needs to be located on the diagonal. As such, only the x co-ordinate is calculated:

$$x_f = \frac{L}{2\sqrt{\epsilon_e}} = \frac{16.751\text{cm}}{2 \times \sqrt{4.4}} = 3.993\text{cm}$$

All of these theoretical results provide a good starting point for simulating both the diagonally fed square patch, and the truncated edge patch. Using Sonnet EM software, the dimension parameters will be entered, and the results can be improved by means of parameter sweeps. The first design, the square patch is shown first, followed by the truncated edge patch in Figure 20:

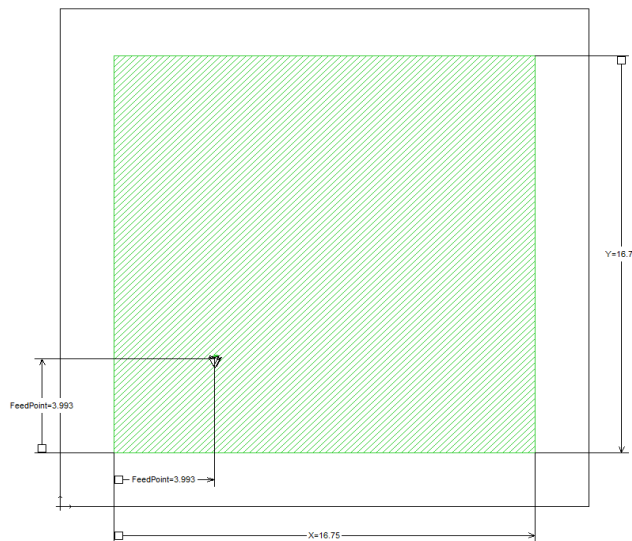


FIGURE 19: PARAMETRIC 2D VIEW OF SQUARE PATCH ANTENNA

The truncated edge patch antenna has a similar design, however the two edges which are not located near to the feed point are truncated. The truncations removed forms a 3 by 3 isosceles

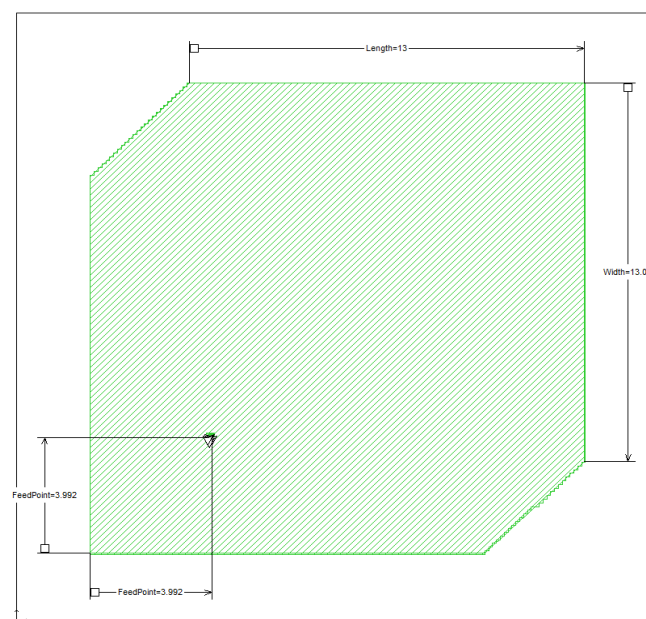


FIGURE 20: 2D PARAMETRIC VIEW OF TRUNCATED EDGE PATCH ANTENNA

triangle. The design is shown below:

By using dimensional parameters, it is relatively straightforward to optimize the various dimensions of the patch antenna. The above two figures provide the basic parameters for the antennas. The results of these, and the improvements made to them are documented in the next section.

5.4.2 INITIAL RESULTS AND OPTIMIZATION

The initial results from both designs of antenna show that there are some slight adjustments that need to be made to the dimensions and the feed points. The first antenna under simulation is the square patch antenna. It is also clear from the input impedance results, and the return loss results, that the feed point needs to be adjusted. Once the correct frequency and a good impedance match is achieved, the gain and efficiency characteristics can be observed using CST MWS.

The square patch antenna results showed that the return loss is only -18.8dB. This is a reasonable result, however the impedance match results show that an improvement can immediately be made by moving the feed point further towards the edge of the patch. Movement in this direction will increase the input impedance, which the lighter trace above shows is currently at 46Ω . Adding to this, the frequency shown here is 430MHz, which is 4MHz too low. This means that the patch dimensions need to be made slightly smaller. This will be the first step, as making the patch smaller will effectively move the feed point further towards the edge. Once the correct frequency is achieved, the impedance match will be considered and changed if needed.

In order to find the correct size, parameter sweeps can be carried out over a range of dimensions. A reduction in the size of the patch by 0.2cm to 16.5cm provided resonance at 434MHz. This also had the effect of moving the feed point closer to the edge. Only a minor adjustment needed to be made to the feed point location to provide close to 50Ohm matching. The feed point needed to move only 0.1cm further towards the edge of the patch, achieving 49Ω matching. The Smith charts are particularly useful in terms of determining how well matched the antenna is. As the curve cross extremely close to the origin, with near zero real and imaginary components for the new feed point at (3.9, 3.9)

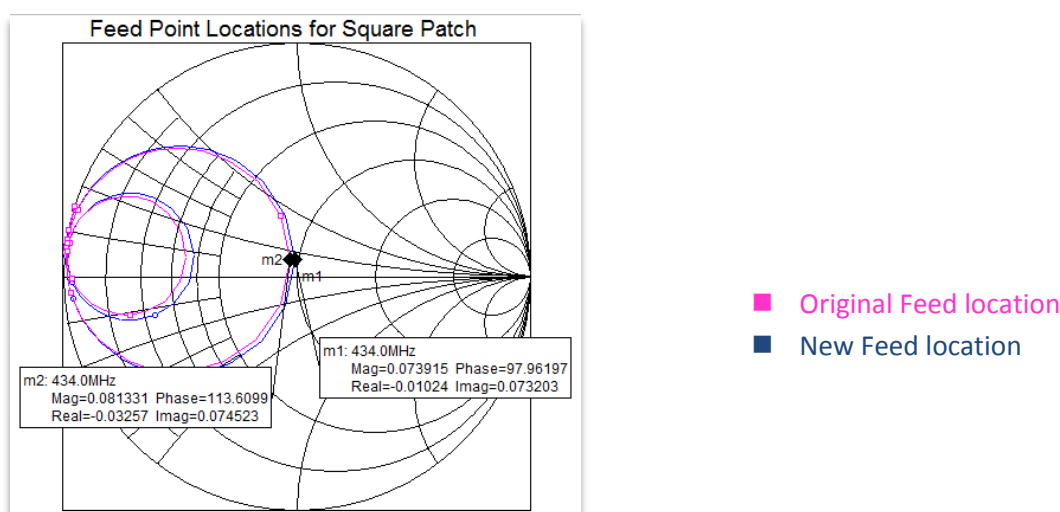


FIGURE 21: SMITH CHART SHOWING ORIGINAL AND NEW FEED LOCATIONS

The results below show the final return loss and impedance for the square patch antenna. Also shown is the -10dB bandwidth.

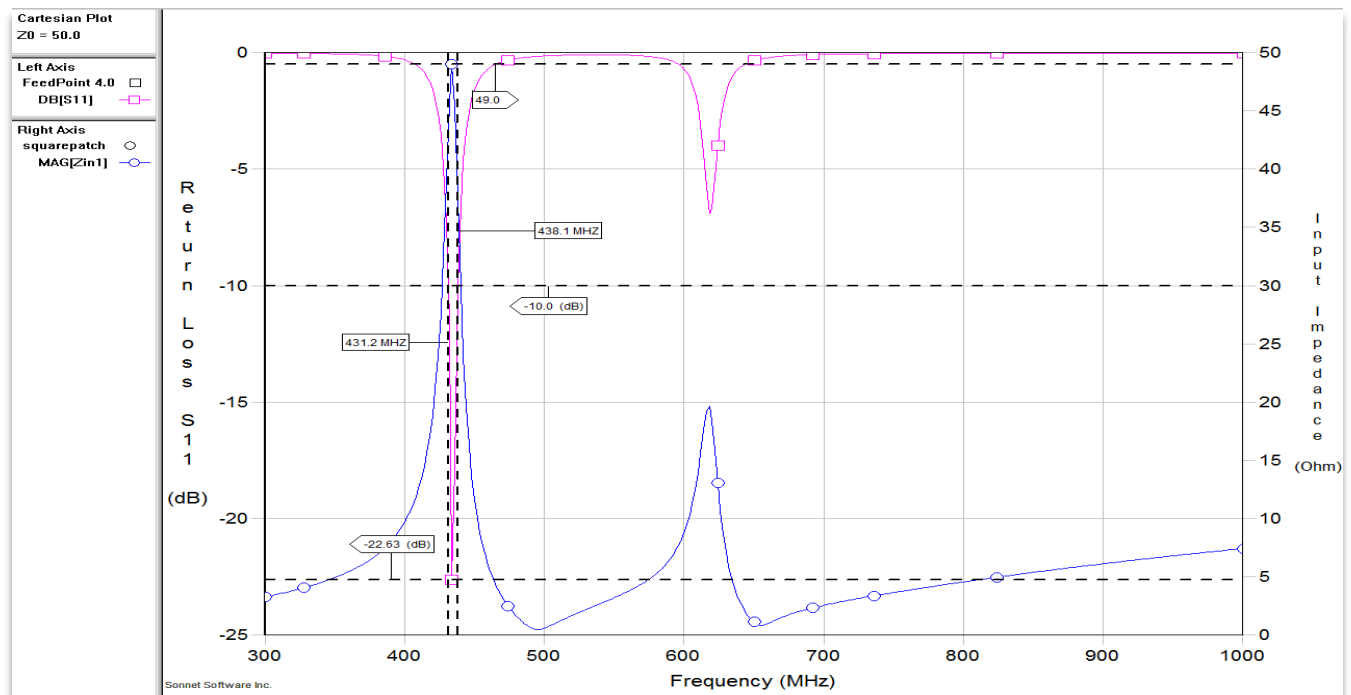


FIGURE 22: IMPROVED RETURN LOSS RESULTS FOR THE SQUARE PATCH ANTENNA

These results show the optimum configuration for the square patch antenna. An important aspect to note is that there is a second band at which this antenna has some resonance at. If the second band was lower than -10dB, then this would become problematic. However, since the second band is less than 25% of the 434MHz band, it is not a problem for this antenna. Adding to this, the second band at 620MHz becomes less significant as the impedance match improves.

Bandwidth is another important factor, particularly for a patch antenna. Patch antenna designs can be typically quite narrowband, and this is the case for this antenna. Taking the bandwidth at the -10dB point shows that this antenna has a bandwidth of 6.9MHz. This equates to approximately 1.6% of the centre frequency. This is quite low, even for a patch antenna, where typical bandwidths are of between 3 and 6% of the centre frequency. Some improvements can be made through material changes and tuning techniques, but these methods will be analysed after one of the two designs is selected.

The square patch antenna can now be compared with a truncated edge square patch antenna as was designed parametrically in Figure 20. The initial results for this antenna already show an improvement over the square patch. This antenna provides a lower return loss, even when matched poorly. The initial results are shown below:

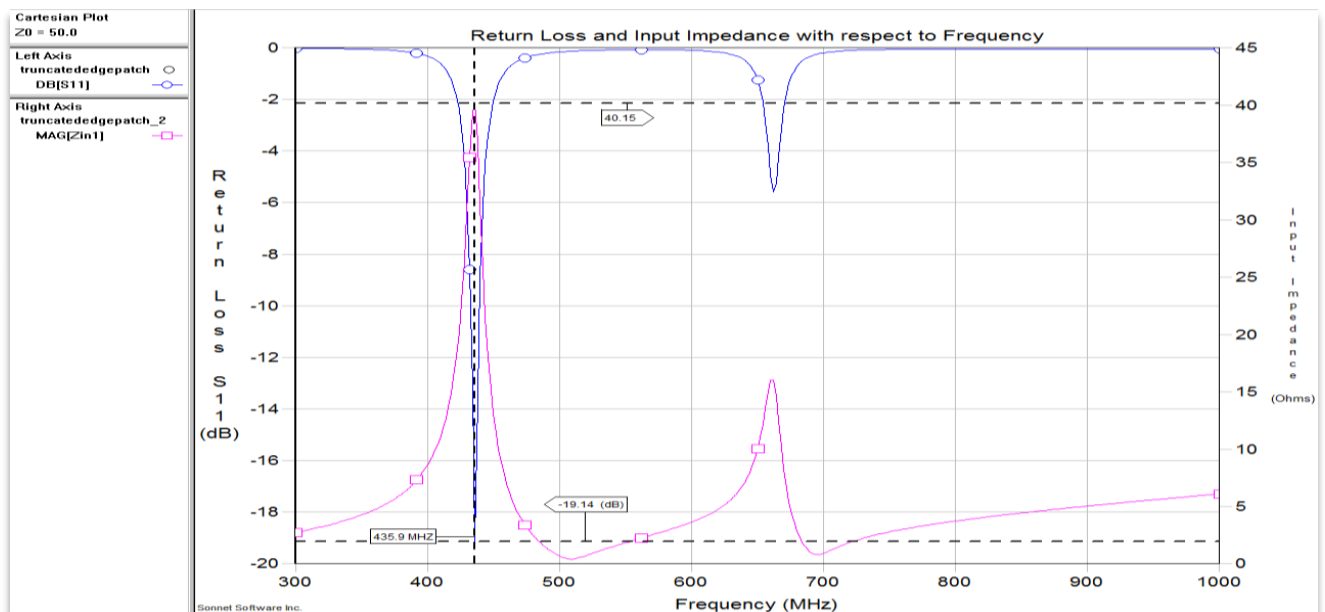
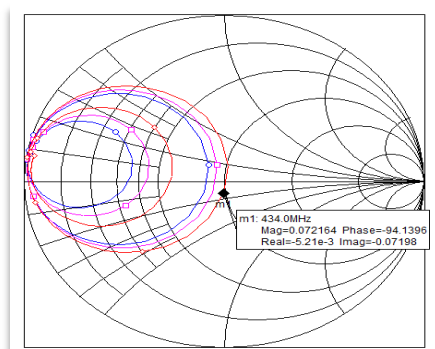


FIGURE 23: INITIAL RETURN LOSS AND IMPEDANCE FOR TRUNCATED EDGE PATCH ANTENNA

These results similar initial return loss values as the square patch antenna did for its initial values, however in this case, the impedance match is worse. This means that it is predictable that this design will provide better return loss results than the square patch when it is matched properly. As was the case with the square patch antenna, this design needs to be tuned slightly to achieve the correct resonant frequency. However, in this case the patch needs to be made slightly larger than its current dimensions, since the frequency is too high. The impedance point needs to be moved further towards the edge also to achieve an impedance match closer to 50Ω . What is also visible from these results is that the second band of operation at 680MHz is already smaller than it was for the optimum square patch results.

Tuning of this antenna is again carried out using parameter sweeps, which showed that the antenna needed to be 0.2cm larger to achieve resonance at 434MHz. By making the antenna larger, it effectively moved the feeding point inwards, thus providing lower impedance. Tuning of the feed point is again critical to ensure that the patch antenna operates efficiently. A large number of feed points were simulated, with each moving closer towards the edge. Some of the locations of the feed points are shown below on the smith chart. Those passing directly through the origin prove the best impedance match:



- Original Feed location
- Feed Location at (3.6, 3.6)
- Feed location at (3.2, 3.2)

FIGURE 24: SMITH CHART SHOWING 3 DIFFERENT FEED POINTS

It is clear to see that as the feed point is moved towards the edge of the patch, the match to 50Ω becomes closer, as shown by the curves moving towards the origin. The feed location of (3.2,3.2) provides a near perfect impedance match to 50Ω . The truncated patch antenna provides a slightly larger bandwidth than that of the square patch, as shown below:

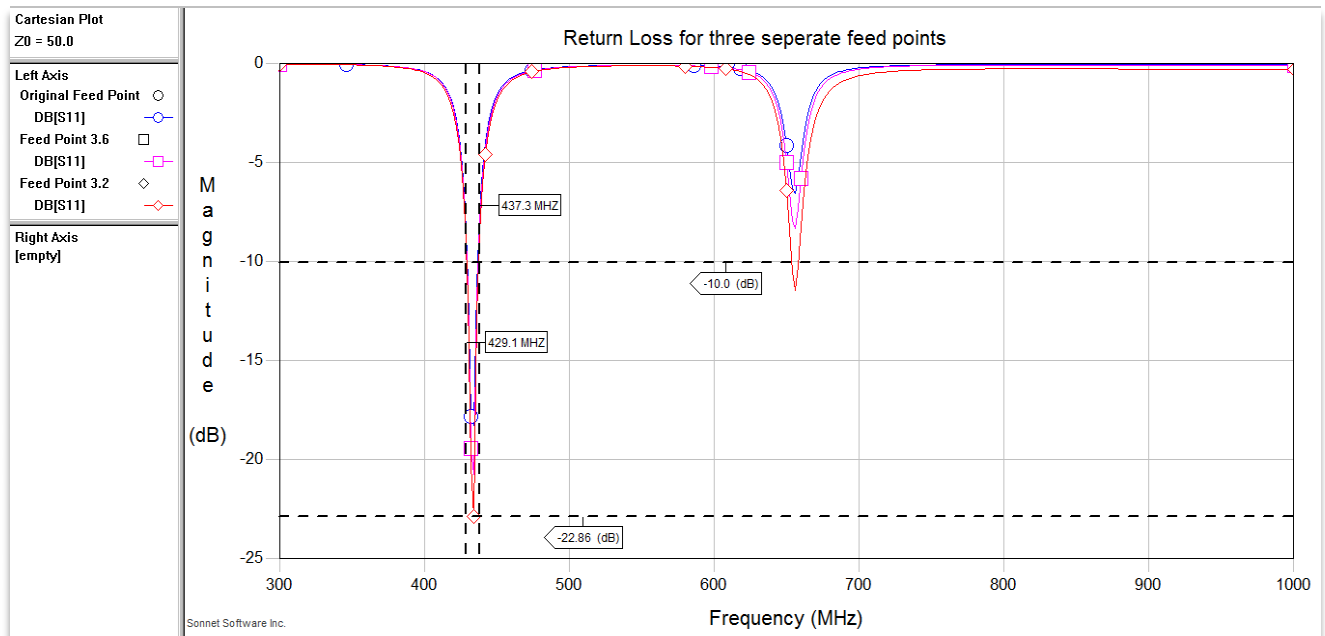


FIGURE 25: RETURN LOSS FOR THREE SEPARATE FEEDPOINTS

As shown in Figure 25, the bandwidth for this antenna is approximately 1.8% of the centre frequency, showing a slight improvement over the square patch antenna. Bandwidth can be improved by adjusting certain design parameters.

Comparing the two antennas in terms of the voltage standing wave ratio shows that the truncated edge patch antenna has a slightly lower VSWR. The ideal result for the VSWR is a value of 1, however both antennas in this case show results slightly above this. This is shown below:

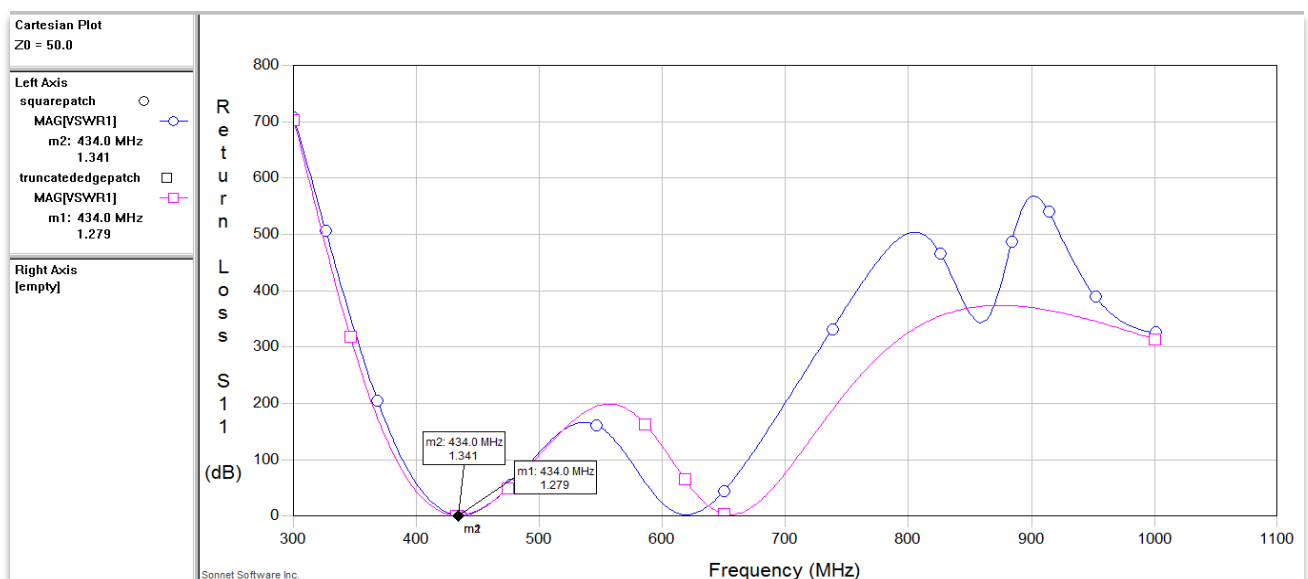


FIGURE 26: VSWR RESULTS FOR TRUNCATED EDGE PATCH AND SQUARE PATCH

The above figure shows that for the truncated corners patch antenna, the return loss is slightly less than that of the square patch antenna. This is consistent with the fact that the return loss is less for the truncated patch. It is also worth noting that other experiments outside of this report have shown similar characteristics in a side-by-side comparison such as this. (Sharm & Gupta, 1983) carries out similar tests at higher frequencies, but showed that the square patch has a slightly higher VSWR than that of the truncated corner patch. The results in this experiment also showed that thinner substrates provided lower VSWR but provided a narrower bandwidth.

Another consideration is the surface current. This is shown below:

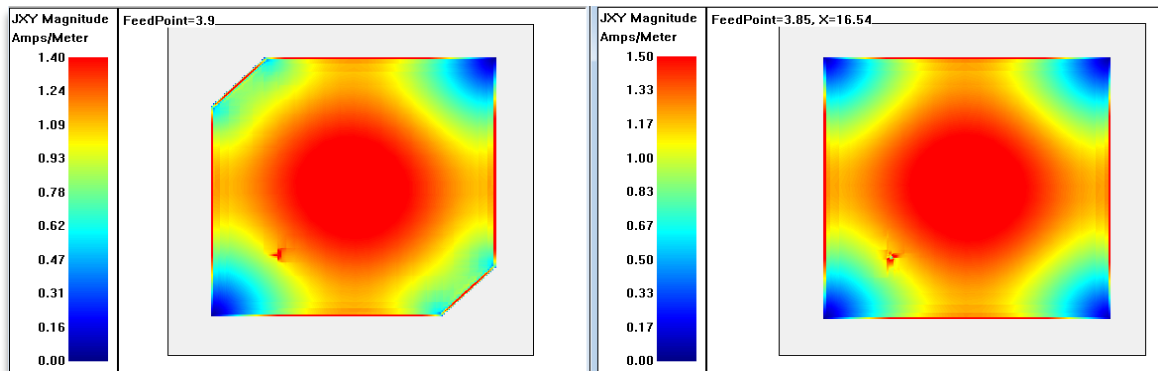


FIGURE 27: SURFACE CURRENT OF BOTH THE SQUARE AND TRUNCATED CORNER ANTENNA

It is clear that both antennas are circularly polarized by observing the current density on the surface of the conductor. However, the corner-truncated patch antenna makes slightly more efficient use of the surface area by removing the corners that have minimal current density. However, there is no large difference between the magnitudes of the surface currents.

A lot of data has been presented here in regard to the two patch antennas. Both are fully optimized with both providing good return loss results. However, on balance, the truncated patch antenna provided better results, both in terms of bandwidth and VSWR. Adding to this, it makes more efficient use of the conducting area than the square patch antenna. The advantages of the corner truncated are further enhanced when looking at external experimental results. The results provided in (Sharm & Gupta, 1983) show that the corner truncated patch provides better axial ratio and a larger axial ratio bandwidth than the square patch antenna. It must also be noted that for both cases, the results are reasonably consistent with those in the aforementioned paper, particularly in terms of the VSWR. As a result, a corner truncated antenna will be constructed using CST Microwave Studio (CST) to view the gain, axial ratio and beam patterns.

5.4.3 DEVELOPMENT OF TRUNCATED CORNER PATCH ANTENNA

The corner truncated patch antenna was constructed now in CST and provided similar results to those in the Sonnet EM software. First step is to consider the current gain, beam pattern and efficiency characteristics. The initial gain and efficiency results for the truncated corner patch antenna are shown below:

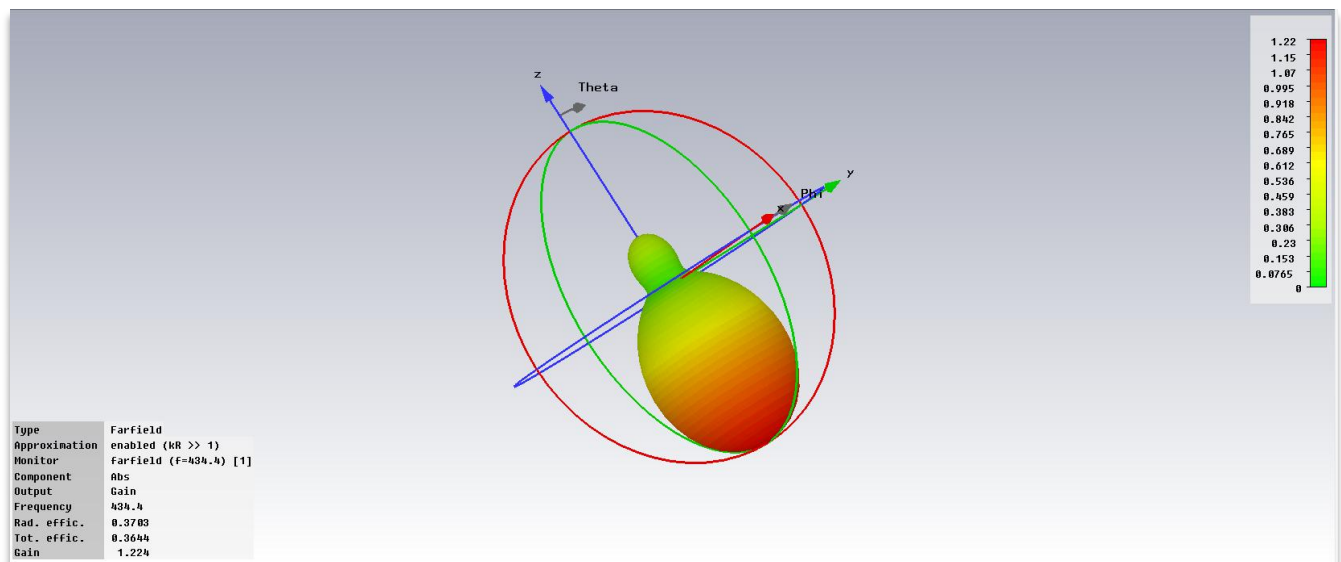


FIGURE 28: INITIAL GAIN AND EFFICIENCY RESULTS FOR CORNER TRUNCATED PATCH

It is noted from the results of this simulation that the efficiency is quite poor, at only 36.4% and the gain is only 1.224dB. It is difficult to determine the exact cause of this, but it most likely due to dielectric constant of FR4. (Yadava, 2011) states that ideally, the substrates used should be thick and have a low dielectric constant. Since using a low dielectric constant is not an option due to the inverse relationship between ϵ and the length of the patch, a thicker patch can be designed. FR4 also has a reasonably high loss tangent compared with higher quality materials. Changing to a higher quality and slightly thicker substrate may improve the efficiency results. Thermoset Microwave Materials (TMM) made by the Rogers Corporation are a range of substrate materials. TMM4 is a material with a dielectric constant of 4.5, which is very close to that of FR4. This means little needs to be done to adjust the dimensions of the antenna. The thickness chosen was 3.175mm.

The results from the TMM4 simulation provided the ideal operation of this patch antenna. The change of substrate gave excellent return loss results with increased bandwidth and efficiency. The combination of the thicker substrate and the lower loss tangent of TMM4 provided efficiency close to 80%. FR4 typically has loss tangent of 0.02, whilst TMM4 has a loss tangent ten times smaller than this at 0.002. The increased quality of this material does however increase the cost of the substrate, and it is unfortunate that Rogers do not supply this material at a low price. However, some of the parameters from this optimum configuration, such as the increased substrate height can be carried forward to provide better gain, bandwidth and efficiency results.

(Garg, et al., 2001) explains how the substrate thickness can have a large impact on a number of parameters of a patch antenna. By having a thick substrate it is possible to increase the radiated power of the antenna, thereby potentially increasing the efficiency. A thicker substrate will also have an impact on the dielectric losses, and this may be a problem in a lossy dielectric. However, the contrasting aspect of this is that by having a thicker substrate, it effectively lowers the dielectric constant. Due to size limitations, it is not possible to have a small dielectric constant which would be beneficial, however by having a thicker substrate, it is possible to get closer to the lower values of dielectric constant which would provide better results.

Since other substrates are unavailable, the design has to be fabricated using FR4. The limits on the thickness of FR4 make it more complex to use a thicker substrate; however it is possible to layer pieces of FR4 to fabricate a thicker substrate. This provided a set of good results which can be taken forward to fabrication. The first result shown below is the return loss for the patch antenna using two sheets of FR4, forming an effective thickness of 0.315mm.

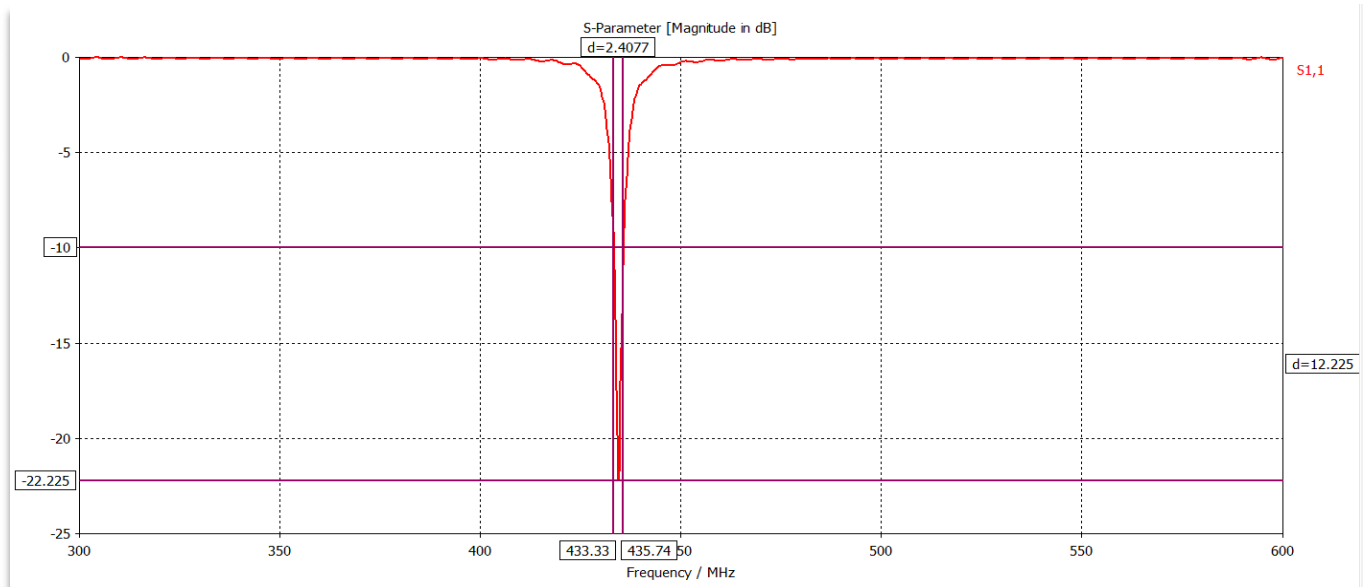


FIGURE 29: RETURN LOSS RESULTS FOR DOUBLE SUBSTRATE PATCH ANTENNA

As the result above shows, there is very little impact on the return loss values when compared to the initial results in Figure 25. This means the substrate thickness has only a minor impact on the matching of the antenna. It must be noted that the feed point needed to be moved closer to the centre of the antenna to provide the above return loss results.

The positive results are carried forward to the gain and efficiency results which both show an improvement:

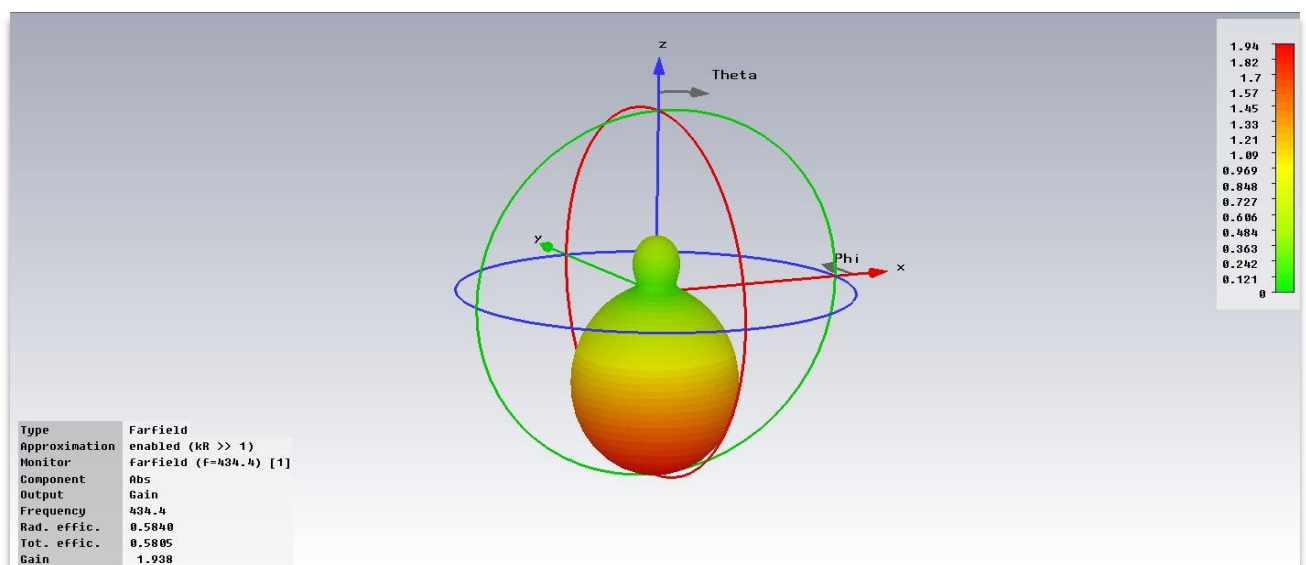


FIGURE 30: GAIN AND EFFICIENCY OF DOUBLE SUBSTRATE PATCH ANTENNA

The results in Figure 30 show the improvement that the two layers of FR4 has on both the efficiency and the gain of the antenna. The efficiency has almost doubled to 58%, whilst the gain has increased from 1.2dB to 1.98dB. This developed design provides an axial ratio of -1.5dB for this antenna. It is noted for reference purposes that this antenna design is Right Hand Circularly Polarized.

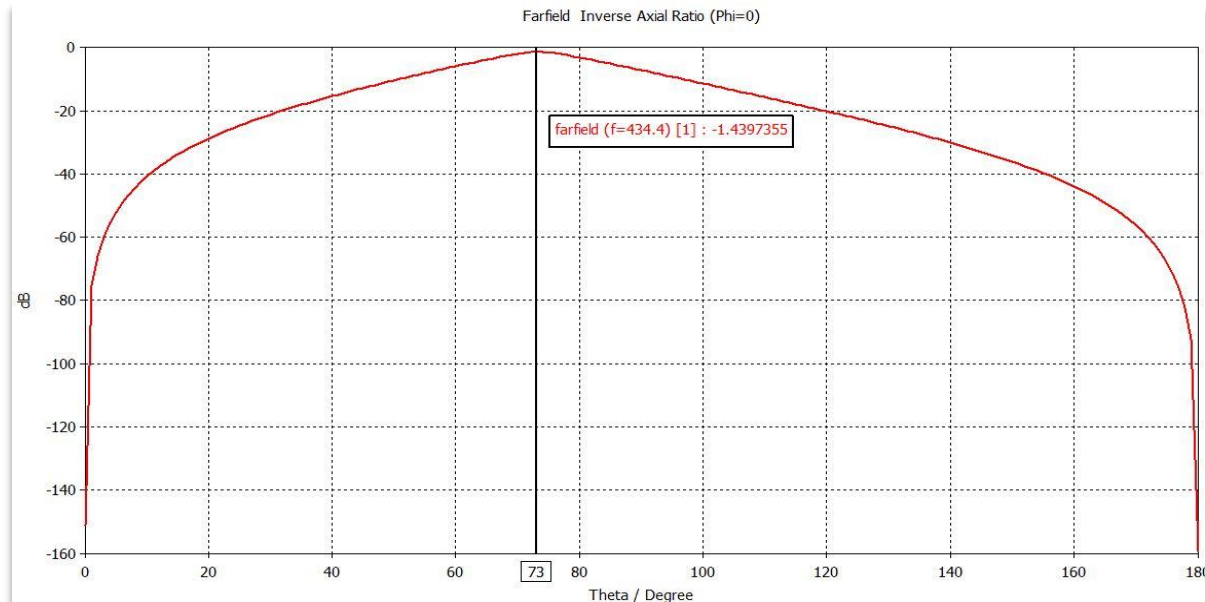


FIGURE 31: AXIAL RATIO FOR DOUBLE SUBSTRATE PATCH ANTENNA

5.4.4 TUNING METHODS

Despite the accuracy of the simulations, it is common that the materials have minor variations leading to differences between the actual fabricated antenna results and those achieved through simulations. The main parameter that needs to be tuned is the centre frequency of the patch antenna. Since the frequency is determined by the size of the antenna and the dielectric constant, adjusting these parameters after fabrication will result in tuning of the centre frequency.

There are a number of different methods for tuning the resonant frequency of a patch antenna. The most common methods for tuning the resonant frequency of patch antennas include tuning stubs and tuning gaps. These are pieces of copper which are included in the design in relevant locations that can be cut down until the correct frequency is achieved. This essentially means that the patch is made so that the resonant frequency is lower than what is desired, and then the stubs are cut down until resonance at the desired frequency is achieved. The inherent disadvantage with this method is that the frequency can only be tuned upwards, since the patch is made larger than required.

The dielectric constant of the substrate can be changed by the use of shorting pins or posts. These provide a level of inductance, thereby changing the permittivity of the substrate itself. (Lee & Luk, 2010). The results presented in the aforementioned reference show that the resonant frequency can be adjusted by up to 300MHz for a small movement in the location of the shorting pin. Although very effective in terms of tuning, it must be noted that this has an effect on the polarization of the antenna.

The third mechanical tuning method that can be used is the use of an air gap between the substrate and the ground plane. Similar to the tuning stubs method, this technique only provides an increase

in the resonant frequency. This is due to the air gap effectively decreasing the relative permittivity of the substrate, thereby, through the inverse relationship between resonant frequency and dielectric constant, providing an increase in the frequency.

The adjustable air gap and tuning stub methods can both be adapted for tuning this particular antenna. Instead of implementing the air gap between the ground plane and substrate, the air gap can be placed between the two layers of substrate. A similar method has been tested previously and also proved positive in terms of the bandwidth of the antenna. (Sharma, et al., 2008). Although the primary method in the referenced paper was to improve the bandwidth, it was also shown that the frequency of the antenna could also be adjusted. Both of these impact positively on the design.

5.4.5 PAYLOAD ANTENNA DEVELOPMENT SUMMARY

The initial equations for determining the size and feed location of the patch antenna were of good accuracy and provided a good starting point for the simulations of the patch antenna. Both Sonnet EM and CST Microwave Studio were used to simulate the antennas, both providing similar results. After the initial simulations in Sonnet, the truncated edge antenna was chosen due to its wider bandwidth and more efficient use of the limited area. The gain and efficiency characteristics were analysed, but were not satisfactory for this antenna design, so further development based on previous experiments and literature was carried out. It was determined that an increase in substrate thickness provided close to twice the efficiency and increased the gain by close to 0.8dBi. The final dimensions determined from simulations will be fabricated and the antenna can be tuned using two pre-planned techniques. The development of the fabricated antenna for the payload is included in Chapter 6 along with the final development and testing of the payload itself and the ground antenna.

5.5 GROUND ANTENNA DESIGN AND SIMULATION

To receive the signals from the payload, it is necessary to have an antenna connected to the receiver. Due to the inherent lack of gain from patch antennas, it is necessary to have a high gain antenna at ground level. The ground antenna can also be considered as a backup tracking system, whereby if the GPS receiver malfunctions or loses signal, but the radio transmitter continues to transmit, a highly directional and high gain antenna will be able to locate the signal. As a proof of this concept, this is in fact a commonly used technique when tracking radio tagged wild animals.

These two specification points limit the design of the antenna to only a few different types of antennas. A parabolic dish antenna would be the ideal choice for this use due to its high directivity and very high gain. However, these types of antennas require accurate fabrication to achieve the optimum results. As such, a slightly simpler high gain antenna can be used.

There are two designs under consideration. The first is the helical antenna which is circularly polarized but provides medium gain. The second antenna being considered is the Yagi antenna, which is typically a high gain, linearly polarized antenna. The Yagi antenna can provide the best gain for a smaller size out of the two antennas and as such, shall be used for the ground antenna.

A standard Yagi antenna is formed of a dipole and a number of directional elements in front of it and a single reflector element to the rear of the dipole. The higher the number of director elements the higher the gain and the directivity. However, more directors mean that a longer antenna needs to be

designed, making the device potentially impractical. Design calculators have been developed to provide an accurate set of dimensions and the relevant predicted gain results. This removes some of the requirement of simulation prior to fabrication; however, some simulations have been carried out to check the results of the design calculators. The design chosen is a 7 element design with 5 directors and a single reflector. The sixth element is the fed dipole. The image below shows both dimensions and the relevant predicted gain from the antenna calculator:

VK5DJ's YAGI CALCULATOR

Yagi design frequency = 434.65 MHz
 Wavelength = 690 mm
 Parasitic elements contacting a square section metal boom 25 mm across.
 Folded dipole mounted same as directors and reflector
 Director/reflector diam = 6 mm
 Radiator diam = 6 mm

REFLECTOR
 354.5 mm long at boom position = 30 mm (IT = 165.0 mm)

RADIATOR
 Single dipole 322.1 mm tip to tip, spaced 138 mm from reflector at boom posn 168 mm (IT = 148.5 mm)
 Folded dipole 348.5 mm tip to tip, spaced 138 mm from reflector at boom posn 168 mm (IT = 161.5 mm)

DIRECTORS

Dir (no.)	Length (mm)	Spaced (mm)	Boom position (mm)	IT (mm)	Gain (dBd)	Gain (dBi)
1	314.6	51.7	219.7	145.0	4.8	6.9
2	311.0	124.2	343.8	143.0	6.5	8.6
3	307.7	148.3	492.1	141.5	7.8	9.9
4	304.6	172.4	664.6	140.0	8.9	11.0
5	301.7	193.1	857.7	138.5	9.8	11.9

COMMENTS
 The abbreviation 'IT' means 'Insert To', it is the construction distance from the element tip to the edge of the boom for through boom mounting
 Spacings measured centre to centre from previous element
 Tolerance for element lengths is +/- 2 mm
 Boom position is the mounting point for each element as measured from the rear of the boom and includes the 30 mm overhang. The total boom length is 888 mm including two overhangs of 30 mm
 The beam's estimated 3dB beamwidth is 52 deg

FIGURE 32: YAGI ANTENNA DESIGN DIMENSIONS

These design dimensions provide an approximate result for the dimensions and the predicted gain and beamwidth results. The antenna was constructed from these parameters in CST, and provided very similar results. These are shown below:

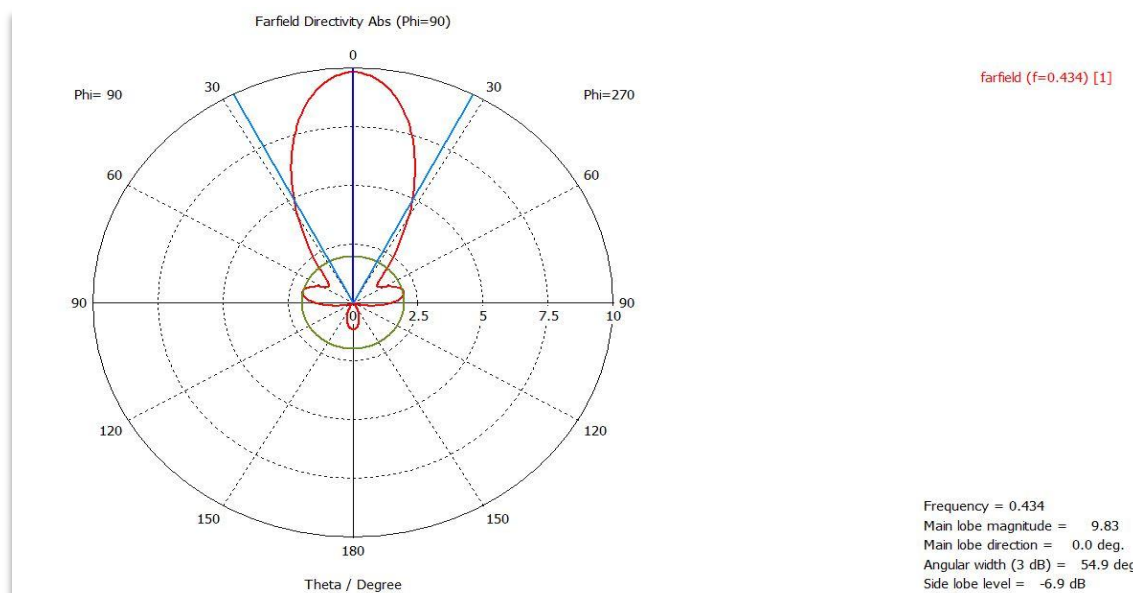


FIGURE 33: FAR-FIELD DIRECTIVITY OF A 7 ELEMENT YAGI WITH 5 DIRECTORS

The last point of Figure 32 states the predicted beamwidth of the antenna will be 52 degrees at 3dB. The simulated results are in agreement with this prediction, showing that the beamwidth will be 54.9 degrees at 434MHz. The gain results are less consistent, providing only 9dBi compared with the predicted 11dBi.

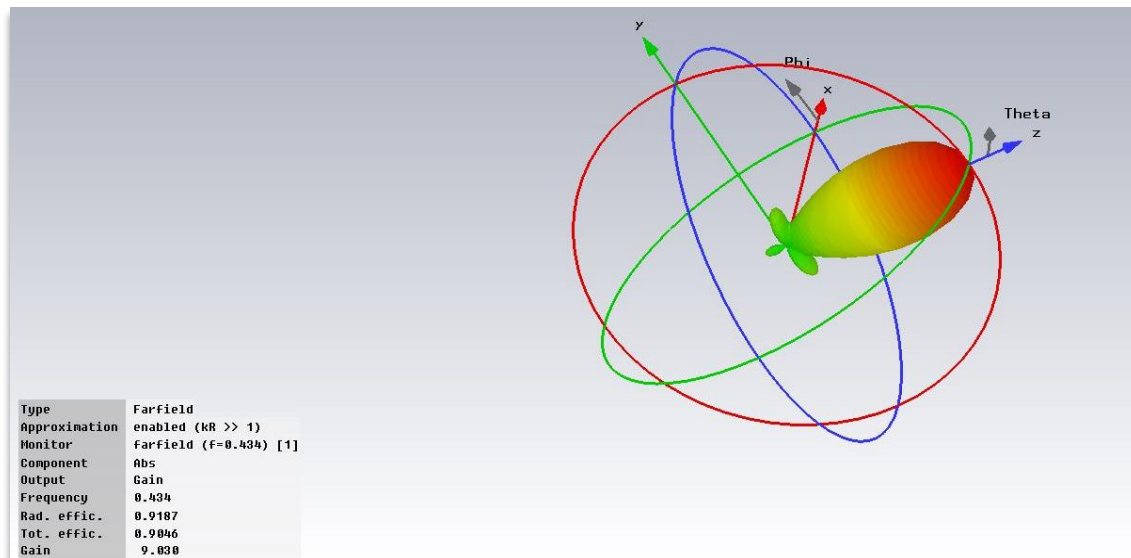


FIGURE 34: FAR FIELD GAIN RESULTS FOR 7 ELEMENT YAGI ANTENNA

Due to the high impedance of copper and brass rods, a matching network or balun is typically required for a Yagi antenna.

With these positive results in mind, due to time constraints it was decided that rather than fabricate the antenna and design the matching network, it would be better to purchase and test a readymade antenna. This would ensure consistent and reliable results.

CHAPTER 6: FABRICATION DEVELOPMENT, TESTING AND ANALYSIS

This section documents the testing of the payload and its relevant systems prior to launch. In order to ensure that the payload will function correctly, both antennas were tested to ensure they were resonating at the correct frequency with minimal reflection. Adding to this, the payload itself was tested to ensure that the components had enough insulation from the cold external temperatures of the upper atmosphere. Prior to launch, it is necessary to run various predictions based on wind data to confirm a launch date. A thorough pre-flight testing regime will ensure that as many problems as possible are eliminated prior to launch.

6.1 PATCH ANTENNA FABRICATION AND TESTING

From the simulations and the development which was carried out in Section 5.4, the patch antenna can be fabricated. The method to implement this is by means of using CAM, or Computer Aided Manufacture. The design is processed from a Gerber file onto two pieces of single sided FR4. Following this, the SMA connectors can be soldered on and the relevant tuning processes carried out. An image of the patch antenna prior to tuning and after tuning can be found in Appendix C

The patch antenna was initially tested using the Vector Network Analyser (VNA) and it was clear that it was resonant at a much higher frequency than anticipated. The initial results were that it was resonant at 558MHz, which was caused by slight contours in the surface of each piece of FR4. These contours gave rise to air gaps at each edge causing a significant frequency shift. By completely closing these air gaps using double sided tape and glue in the relevant areas, it was possible to lower the resonant frequency. Although it was originally planned that the method for tuning the resonant

frequency would be more accurate, a trial and error method became the method of choice. Using this method, the patch could be tuned to the correct frequency.

Despite tuning the air gap to the correct size, the impedance match was considerably harder to adjust. Initial results showed that the impedance match at 434MHz was incorrect. The result of this was that the second harmonic frequency was the dominant frequency. Movement of the feed point allowed the impedance match to improve, allowing the patch to be resonant in the fundamental mode. Despite the improvement made here, the feed point was not optimized since the impedance was now too high.

To tune the impedance without making further modifications to the feed point itself, copper tape was placed around the whole patch, which effectively moved the feed point closer to the centre of the patch. However, this had a negative impact on the results for the resonant frequency, and as such, further tuning needed to be carried out by adjusting the air gap between the two substrates. By making these tuning adjustments and establishing a correct feeding point, it was possible to provide results which are close to the simulated results presented in Section 5.4.3. These are presented below:

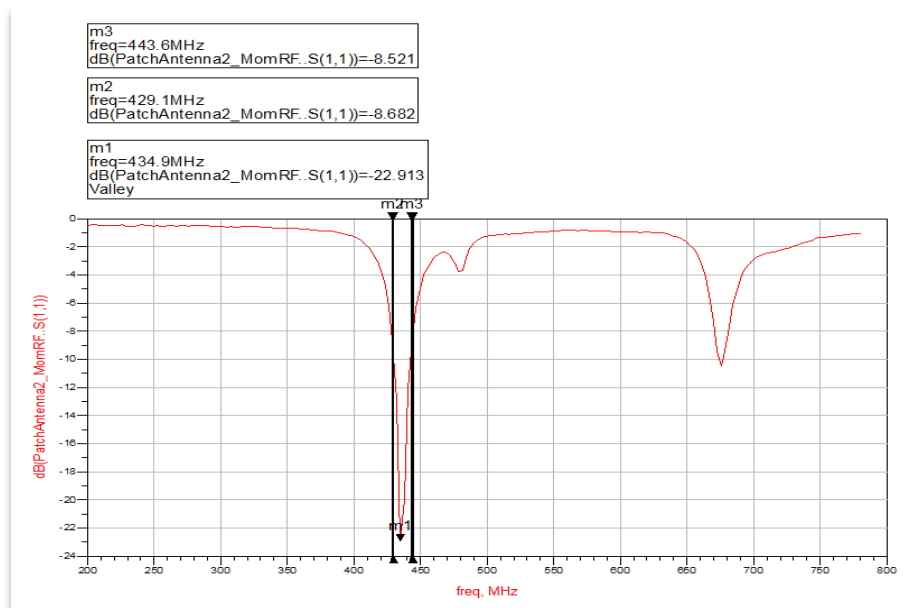


FIGURE 35: FINAL PATCH ANTENNA RETURN LOSS RESULTS

Although the bandwidth in the above graph is taken at -8dB, this can be extrapolated downwards and taken to be approximately 13MHz. This represents 3% of the centre frequency. This is an improvement over the simulated results, and can be attributed to the air gap between the two substrates. This is consistent with the results presented in (Sharma, et al., 2008) which showed an increase of bandwidth with an accurately tuned air gap. The return loss is similar to that present in the simulations and this is supported by an impedance match shown on the VNA of 48Ω. The patch could then be connected to the transmitter, which showed a marginal noise floor improvement when connected.

It is common that an antenna is placed in an anechoic chamber; however, due to time constraints this was not feasible. This is an area for future improvements and development.

6.2 YAGI ANTENNA TEST

The Yagi antenna was chosen for its high directivity and gain characteristics. Due to time constraints, rather than construct a Yagi antenna, an antenna was purchased which would provide more reliable and consistent results in the space of time available. The antenna purchased was in fact the same design that was simulated, except this made use of a folded dipole.

The device purchased specified 9dBi gain for a 7 element design, with a VSWR of 1.5:1. (LPRS, 2012). This shows consistency between the simulation results provided in Figure 34 and the specified characteristics of the purchased Yagi Antenna. To ensure that the Yagi antenna worked within the specified frequencies, it was tested using the Vector Network Analyser (VNA). The results are shown below:

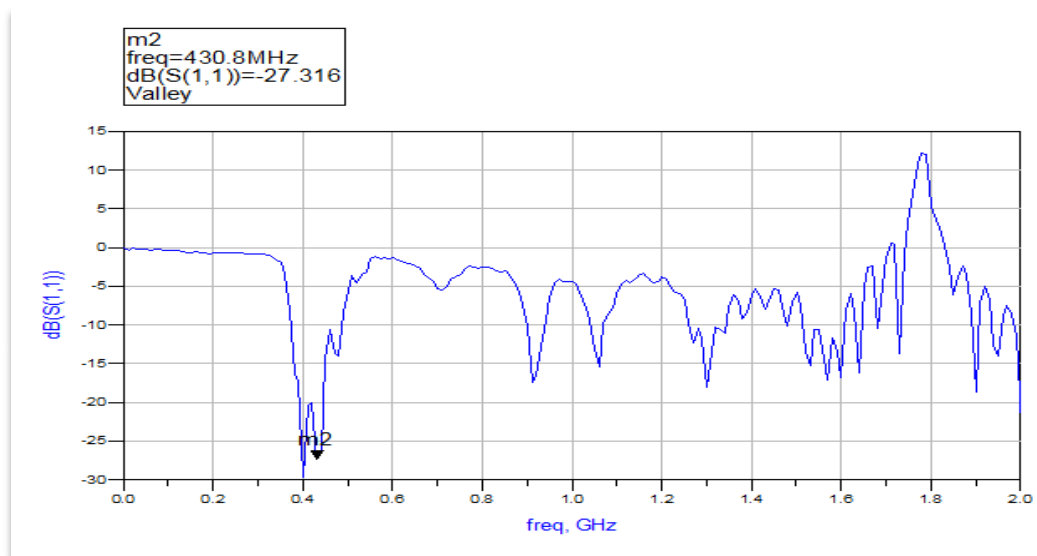


FIGURE 36: RETURN LOSS TEST RESULTS FOR YAGI ANTENNA

The results above show that the antenna has a bandwidth within the specified range on the data sheet. At 434MHz the return loss was recorded at -25dB. This represents a VSWR of 1.12:1. This is better than the specified results. This is most likely because the VSWR in the data sheet is an average over the bandwidth. This shows that the antenna will be able to receive the signal and very little will be lost through mismatch.

6.3 PAYLOAD TEMPERATURE TESTS

The upper atmosphere has a number of temperature changes and it is necessary to ensure that the payload performs consistently at each. The testing facility in this case was to use a regular home freezer, with a minimum temperature of -17°C. Although temperatures during the decent have been recorded at -45°C, this is only experienced for a very short time due to the rapid ascent rate at these altitudes. The test provided an indication of how three aspects of the payload performed. The first was to test the internal temperature variation over a large temperature range. The second was to visualize the maximum temperature variation and finally, the reliability of the transmission system could be tested over an hour long period. The raw data for the temperature and humidity is available in the supporting material however, it is summarised in the graph below:

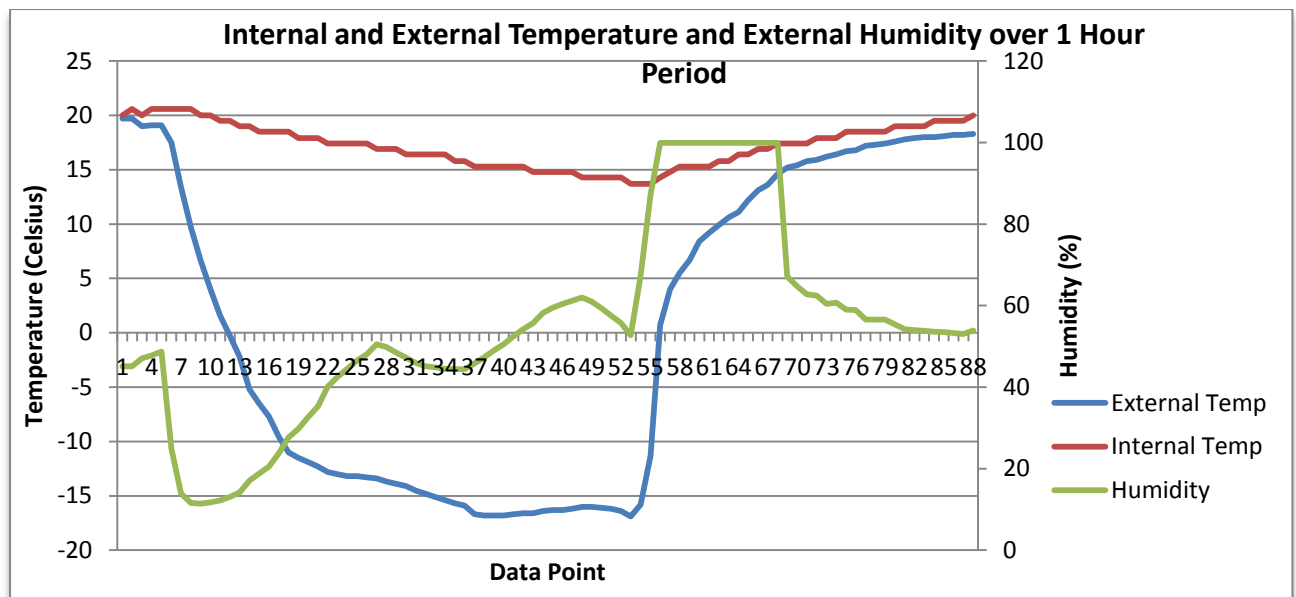


FIGURE 37: INTERNAL AND EXTERNAL TEMPERATURE TEST

From these results, it is clear to see two characteristics of the payload's temperature sensors. The first is that the internal temperature lags considerably behind the external temperature, showing a 5 degree variation whilst the external temperature has close to an 80 degree range. Another observation is that the temperature of the payload is possibly altering the external results due to the mounting position. This may be a minor variation but it should be factored into the accuracy of the results. It is also shown that the humidity sensor reads 100% when condensation forms on the sensor itself.

Further to the actual temperature results, it was observed that there are minor shifts in the transmission frequencies, however, the bandwidth of both antennas were sufficient to prevent loss of transmission. Finally, the number of fully received data strings proved to be sufficient to provide a good range of data over a 1 hour period.

With these results in mind, it can be said that the payload has undergone reasonably extensive testing, meaning that it is now ready to be launched. The procedure, results and the analysis of these is contained in the next chapter.

CHAPTER 7: IN-FLIGHT RESULTS ANALYSIS

7.1 GPS TRACKING/FLIGHT

The balloon and payload were launched on the 20th April 2013 and transmitted data across the UK and transmission was picked up as far afield as France. This allowed reliable tracking and reception of the data. The balloon flight lasted approximately 2 hours. The altitude reached was 23,000m which exceeded the 60000 feet target objective by more than 15000 feet. The network of tracking stations across the UK allowed tracking to continue despite the tracking laptop running out of battery. The Yagi antenna performed well in tracking up until this point, after which it was difficult to receive due to the high directivity and the unknown location. In retrospect, a less directional

antenna would have been better for receiving the data, since most tracking stations tend to use an omni-directional monopole or dipole antenna. The Yagi antenna has its advantages in terms of locating the payload upon landing however.

Throughout the flight the patch antenna performed very well, transmitting consistent data with enough power to receive even at 77 000ft. The graph of the altitude tracking is shown below:

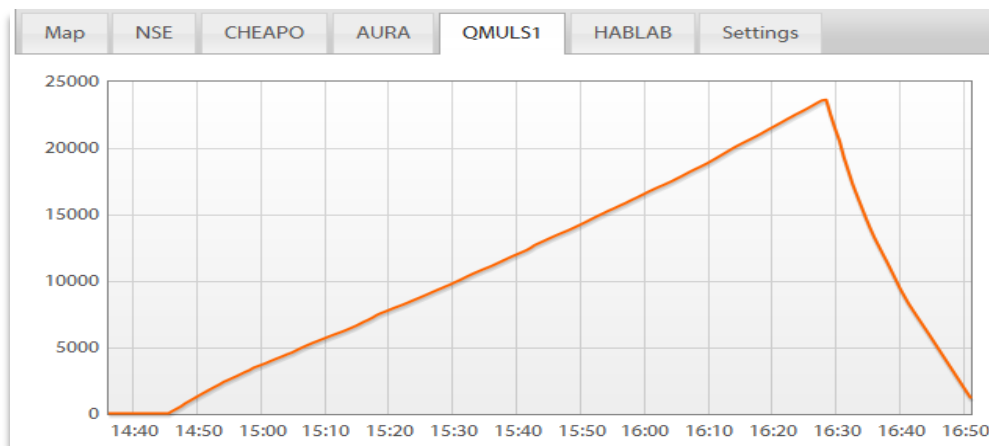


FIGURE 38: REAL-TIME TRACKING OF THE ALTITUDE OF THE BALLOON

The above graph shows the data as it was received during the flight. The last recorded position was at 1km of altitude, at which point communication was lost. Due to a parachute malfunction, the payload fell at a higher than expected rate of descent, and as such the transmitter board shattered on impact. However the last known position was used to extrapolate an approximate location and the payload was retrieved with the visual data still intact. A map of the route is also included in Appendix C. There are improvements that can be made to both the tracking systems and the overall payload structure; however, it does appear that the structure was sufficient to protect the equipment from significant damage. Images of the payload upon recovery are also included in the supporting material.

7.2 SOLAR AND TEMPERATURE DATA

There were two objectives involved in the data capturing systems. The first was to take data readings of the current atmospheric conditions by measurement of the temperature and humidity. The second was to determine if a solar panel rotation system is necessary to maintain optimum solar panel voltage. Both of these objectives can be considered to be completed from observing the data at the end of the balloon flight.

The first data set to be considered is the solar panel voltage data. This data was more of an experimental nature, and was collected to find out the effect of rotating a solar panel to its presumed optimum position to achieve the maximum voltage. The results provide show the result of this experiment.

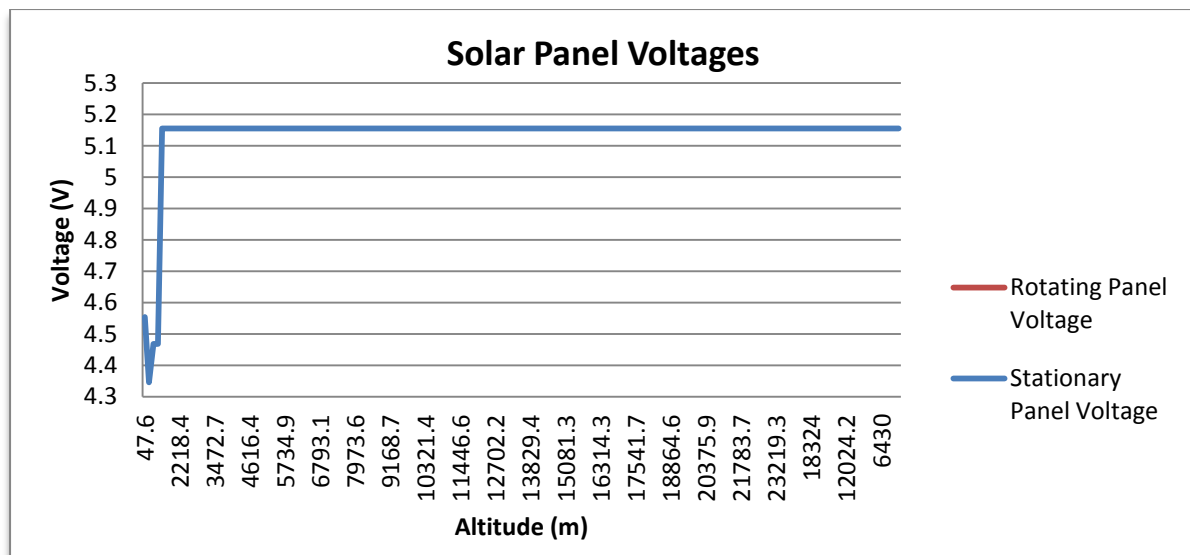


FIGURE 39: SOLAR PANEL VOLTAGES

It is evident from the results above that the rotating panel offered no improvement on the maximum possible voltage. Both panels were constantly at 5 volts for the duration of the balloon flight. This became increasingly predictable throughout the testing phases, in which differences were only noted from highly directed light sources in a relatively low light environment. In an outdoor environment where light is relatively constant at all altitudes, there was very little need for this. Perhaps more sensitive measurements or larger solar panels would have been impacted more by the angle variations; however the size constraints restrict this test somewhat.

The next set of test results are the temperature test results. These results were recorded from both the internal and external temperature sensors, and can be compared with previous experimental data from balloon flights.

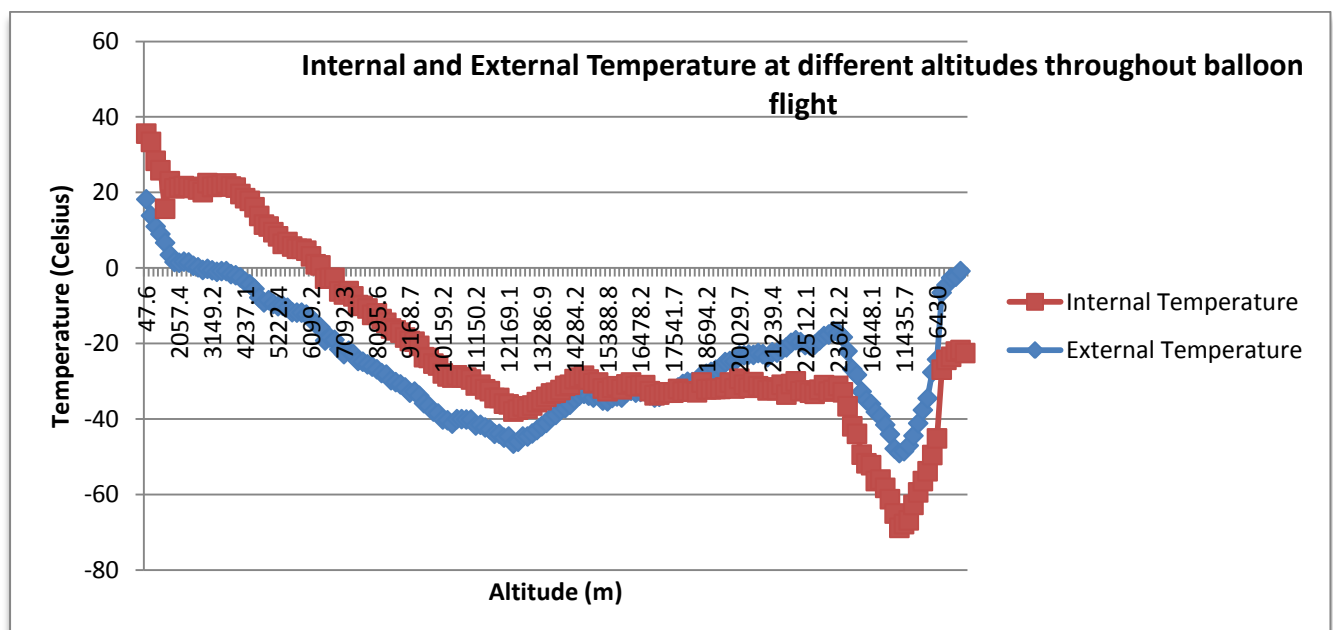


FIGURE 40: TEMPERATURE CHANGES INTERNALLY AND EXTERNALLY THROUGHOUT THE BALLOON FLIGHT

This data shows both the internal and external temperature measurements that were recorded throughout the balloon flight. This can be compared with the graph in Figure 1 and the initial research that was presented in Section 2.1. It is known that above the 10 mile reach of the troposphere, the stratosphere is subject to a temperature inversion. This is caused by ultraviolet rays from the Sun, giving rise to a temperature increase. (Allen, 1983). This factor can be clearly seen in the blue trace of Figure 39

After the initial launch, the temperature falls in a relatively linear fashion towards -42°C . From the 13km mark, the temperature begins to increase from its minimum value and remains with a positive gradient until the burst altitude of approximately 23,000m. At this point, the air flow combined with the re-entering of the cold regions of the atmosphere drops the temperature rapidly, before it rises again as the payload nears the ground. The last recorded temperature reading was at 1km high, and was -0.8°C . The reason that the temperature is lower than it was at this point on the ascent is due to the payload affecting the temperature that the sensor is measuring. Figure 1 showed very similar results to these. Both showed the decrease in temperature, however it is clear that the temperature of this flight decreased further than those presented in (Larson, et al., 2009). Adding to this, the temperature increase could be seen at lower altitudes which can possibly be attributed to the cloudless day and possibly even changes in the atmosphere over the last 4 years since those results were collected.

Another observation which can be seen in this graph is that the insulation on the payload was reasonably sufficient. The red trace, which represents the internal temperature in Figure 40, constantly lags behind the external temperature. However, in contrast to the testing results, the temperature fell rapidly over the descent. This is most likely caused by the air movement over and through the payload.

Humidity data was also taken throughout the balloon flight and is shown below:

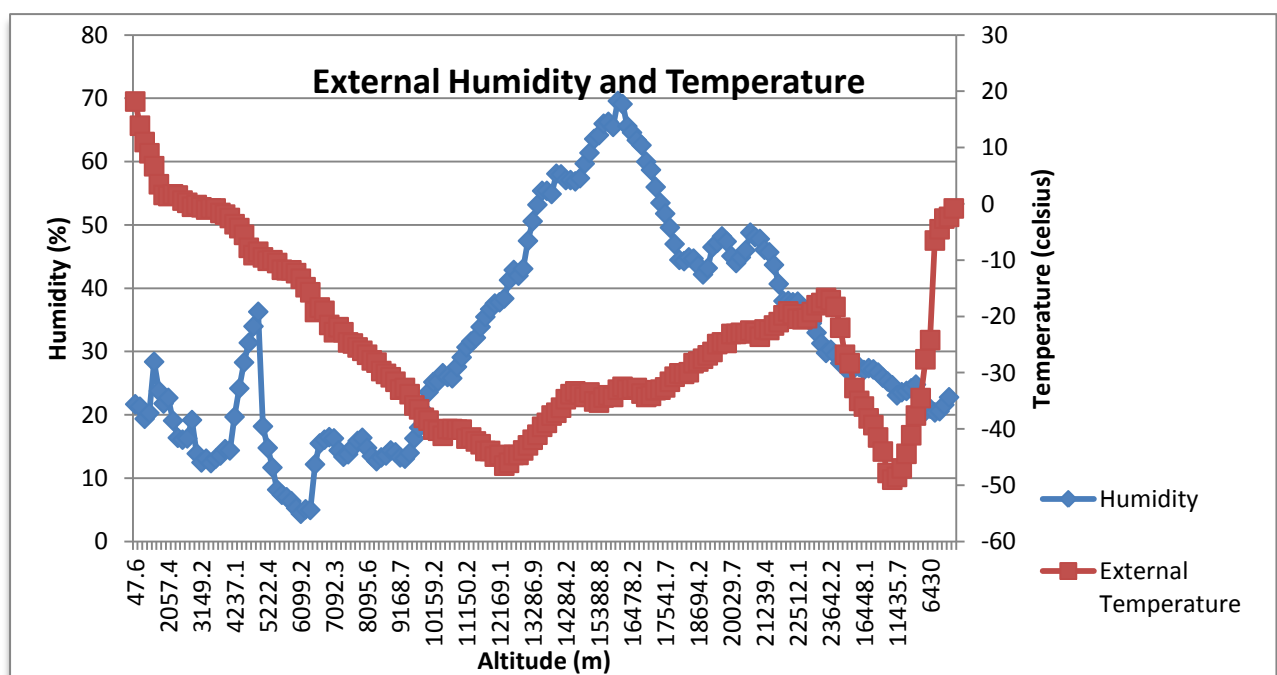


FIGURE 41: EXTERNAL HUMIDITY AND TEMPERATURE THROUGHOUT BALLOON FLIGHT

There is little data available for humidity of the atmosphere, since it changes daily with cloud cover. However, these results do make sense. The initial spike between 4000 and 5000m is most likely to be considered as some low cloud cover or moisture that formed during the ascent. As the temperature began to increase through the inversion layer, the humidity also did so. In these areas, the visual data shows areas of thin cloud that the balloon passes through.

7.3 FURTHER OBSERVATIONS

Other observations of the performance of the system can be seen in the visual footage and the experience of tracking the balloon. The signal strength was sufficient to track the balloon throughout the whole flight and this is shown by the data recorded. The Yagi antenna performed well, as long as it was pointed in the appropriate direction. The receiver stations across the UK tracked the balloon using Omni-directional antennas, which appeared to perform more consistently than the Yagi antenna. The patch antenna performed extremely well and had sufficient gain to transmit the data.

Structurally the payload was sound. Upon impact the equipment was not damaged beyond repair and the data was available. The transmission system with the upload link allowed the data to be stored away from the balloon, meaning the data is available without recovery of the balloon. The parachute needs modification as some knots became loose and the rate of descent was too high, which contributed to the damage sustained upon landing.

CHAPTER 8: CONCLUSION

8.1 DISCUSSION AND IMPLICATIONS

Despite the relatively low cost of the project, which has totaled to £200 including the helium and all components, the payload was able to reach the middle of stratosphere and return intact with the data being transmitted throughout the whole flight. The objectives of the project have largely been completed through detailed research into the development of a transceiver system which included the development of a circularly polarized patch antenna. This formed the core elements of the payload and contributed to the success of the project.

The use of a circularly polarized patch antenna for a High Altitude Balloon is uncommon, however, the design research, implementation and testing of the patch antenna allowed an effective design to be developed. In doing so, comparisons between this work and other experimental work were drawn up. The tuning methods outlined in (Sharma, et al., 2008) proved key to the development of the patch antenna, particularly in terms of the bandwidth and the resonant frequency. Through the design of this antenna, various methods were critically researched, and certain choices were made based on previous work. These choices were simulated and the limitations of the design became apparent, particularly in terms of the bandwidth and the gain. Improvements were made to the design through research, and simulated tests and the final antenna provided good results. The fabrication and tuning of the antenna showed how the air gap between the substrates has a major impact on the results, shifting the resonant frequency by 100MHz for a 1mm air gap. However, the design was successful and transmitted the data across the country throughout the balloon flight. The design is particularly effective due to its planar nature.

Further to this, the temperature data has showed consistency between these results and others such as those in the first figure of this report. The flight has completed the objective of showing the current condition of the atmosphere and has also provided a good starting point for further development. However, it must be noted that potential inaccuracies affected the temperature readings due to the payload itself distorting the external temperature measurements.

Although the results for the solar panel tilting experiment may seem negative at first, the objective was to only find out if this system was necessary. This question has been answered categorically by the results presented in Chapter 7.

Despite the success of the tracking system, some elements of the payload and tracking system did not perform as well as expected. The Yagi antenna proved to be too directional to track the balloon flight at extremely high altitudes, and as such, trackers across the UK had to be relied upon. Further to this, the parachute malfunction during flight resulted in the damage of the payload upon landing. However, despite these failures, the mechanisms put in place, including the overall structure of the payload allowed the equipment to be recovered. The tests carried out prior to the launch of the payload ensured that as many problems as possible were eliminated prior to launch.

8.2 FUTURE DEVELOPMENT

Throughout the development process of the payload, various ideas and experiments were suggested but time constraints and the current design restricted the implementation of these. Adding to this, the in-flight results highlight some areas for improvement and for future development.

8.2.1 IMPROVEMENTS TO THE PROJECT

In its current state, the project achieved the objectives set out in the first chapter of this report. However, there are some areas which need to be improved. The first area is that of the design of the parachute. More care needs to be taken in the construction of the parachute to minimize the risk of entanglement and parachute failure. In doing so, this will protect the payload upon landing, and will ensure that little or no damage is sustained. Further to this, a different ground antenna system should be used, since the one designed was too directional to provide reliable data transmission. However, the Yagi antenna would prove effective in locating the payload if the GPS data was unavailable. Lastly, changes to the location of the external temperature sensor would improve the accuracy of the results taken.

8.2.2 FUTURE DEVELOPMENT AND EXPERIMENTATION

In this section, the areas which can be developed further to achieve the maximum possible results will be detailed. Research into the regulatory requirements for the frequency system highlighted a second band of frequency at 868MHz. This band is more complex to work with, however it can provide higher baud rates, and higher power levels.

Other experimentation that can be carried on the payload could include oxygen and carbon dioxide sensors, pollution measurement devices and further power generation experimentation.

Development of the patch antenna could include different materials for construction, including the aforementioned TMM4 material.

The SDR dongle proved to be satisfactory in its operation, however, it is clear that this system can be further developed to lower the noise floor, and improve the signal to noise ratio. One more complex suggestion for future development in this field is to use a moveable antenna on the ground, which can directly point at the payload at all times providing more reliable data transmission.

8.3 CONCLUSIONS

This report documents the full design procedure, the research and the development involved to launch a payload into the stratosphere via a High Altitude Balloon. A transceiver system was developed to satisfy the objective of transmitting the data reliably across the distance from the payload to the ground. The patch antenna is a new design which has not been tested in this application, and the device performed well during the flight and in the pre-flight testing. This system also took a number of experimental readings throughout the flight and these have been shown to be consistent with previous data. Experiments into photo-voltaic panels showed that there is limited need for a solar tilting mechanism at high altitudes. Overall the objectives set out in the initial chapters, to explore the upper regions of the Earth's atmosphere have been achieved and the data has been collected through the systems designed in this report.

This project provides the basis for further high altitude experimentation on a limited budget and it is hoped that future projects will build on the fundamental work included in this report. New antenna systems can be developed and other frequencies and transmission technologies can be used to achieve altitudes which become closer and closer to the outer reaches of the atmosphere.

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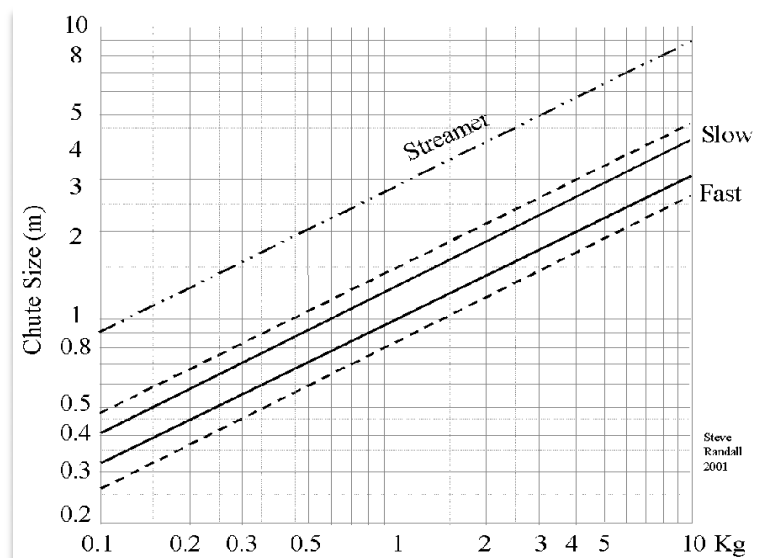
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APPENDIX A

Parachute Sizing Chart sourced from (UK High Altitude Society, 2008)



APPENDIX B

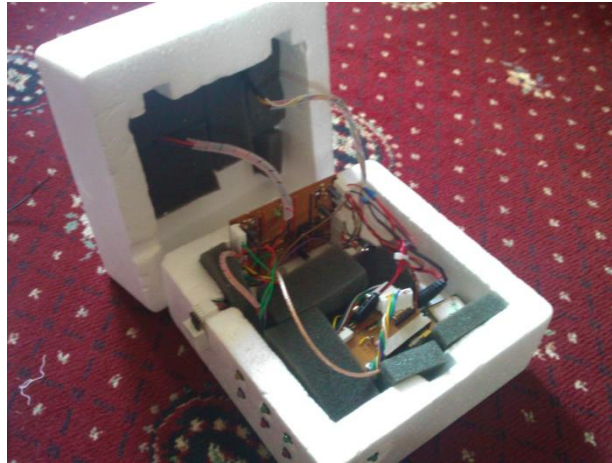
Patch antenna and Yagi antenna Images.



APPENDIX C

Images of the final payload





APPENDIX D

Map of the route taken by the balloon on 20th April 2013

